Urinary Tract & P. A. Wall

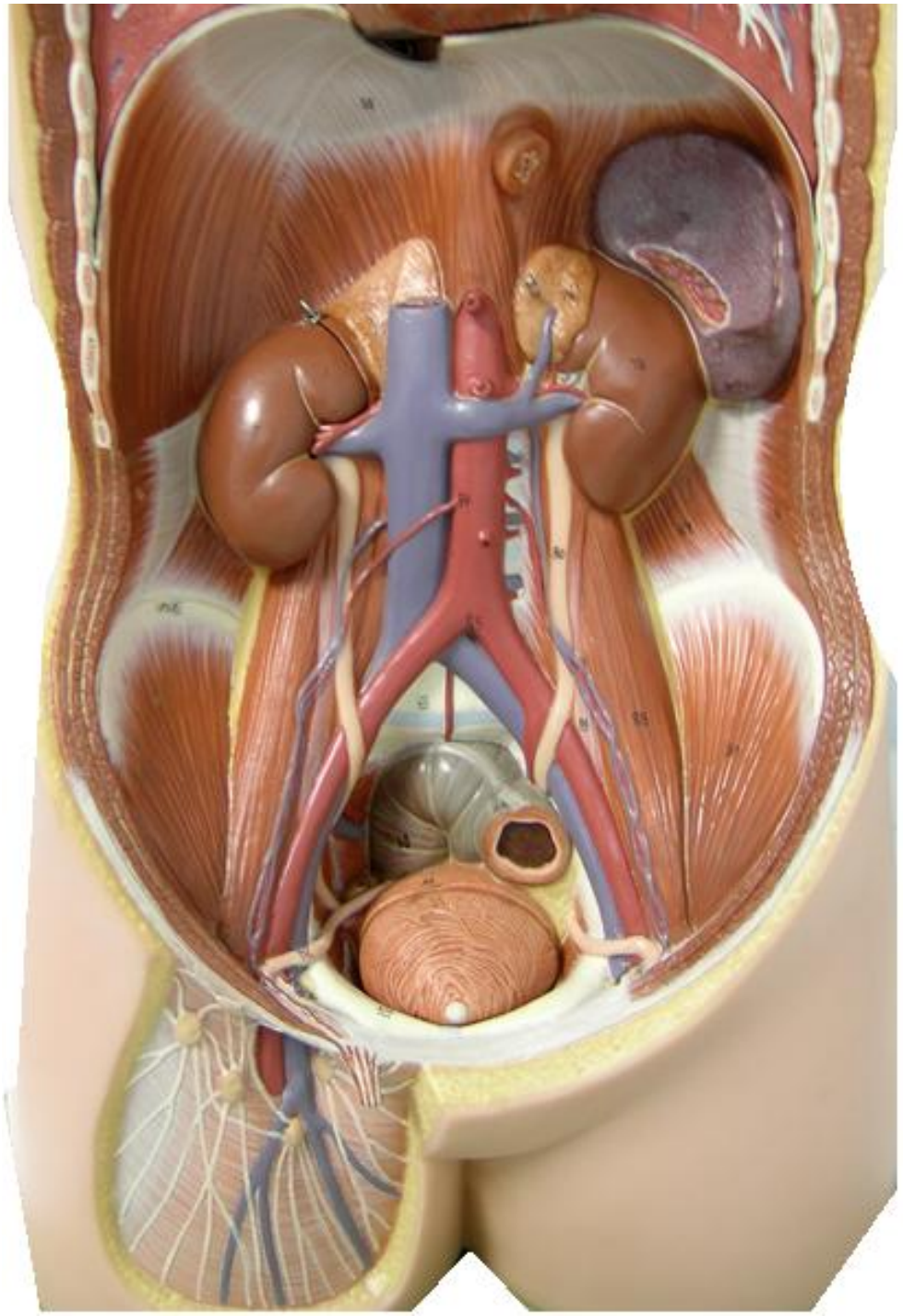
By
Dr. Faris Muhammed
18.10.2026 Sunday

1. Definition

The posterior abdominal wall is the muscular and skeletal wall forming the posterior boundary of the abdominal cavity.

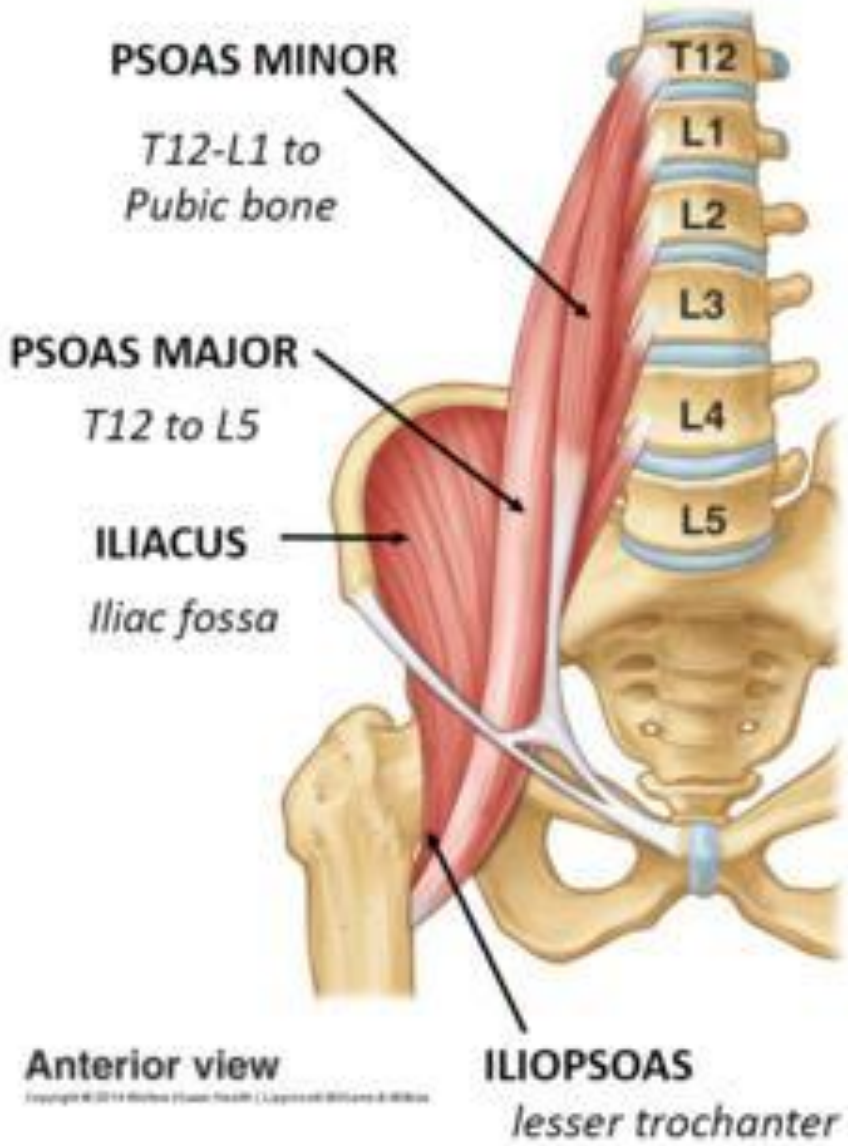
2. Boundaries

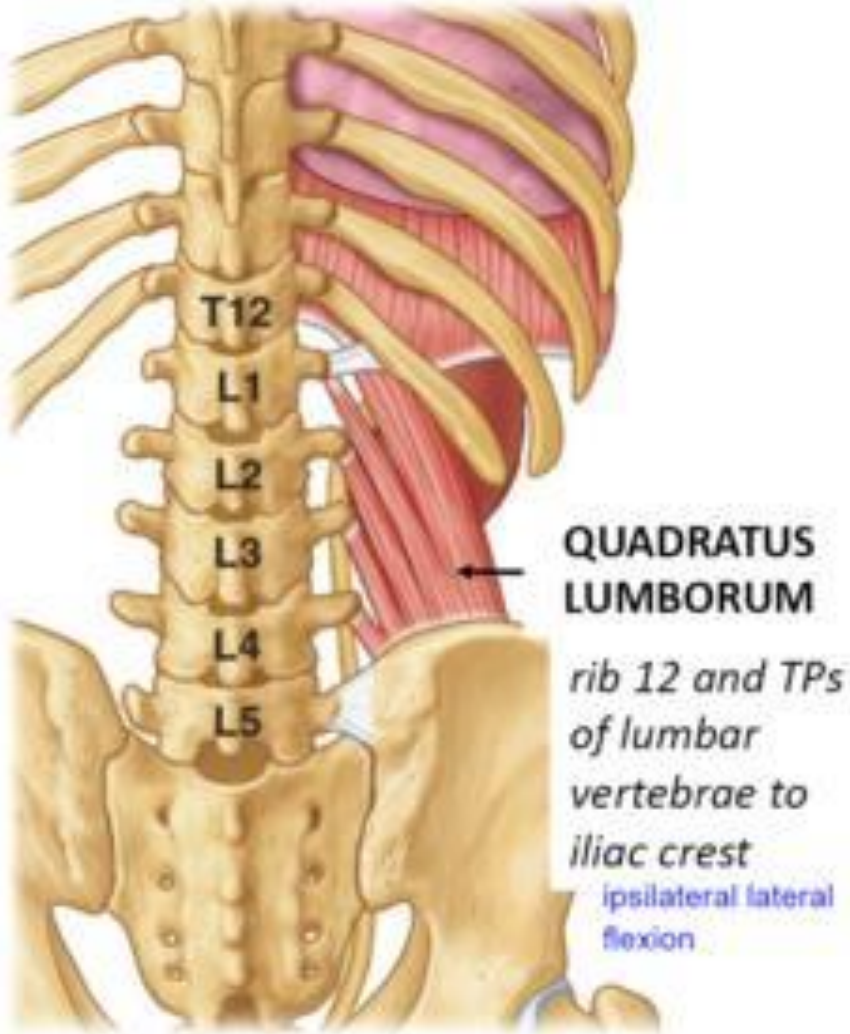
- Superior: Diaphragm.
- Inferior: Pelvic brim.
- Anterior: Abdominal cavity and its contents.
- Posterior: Lumbar vertebrae and posterior abdominal muscles.



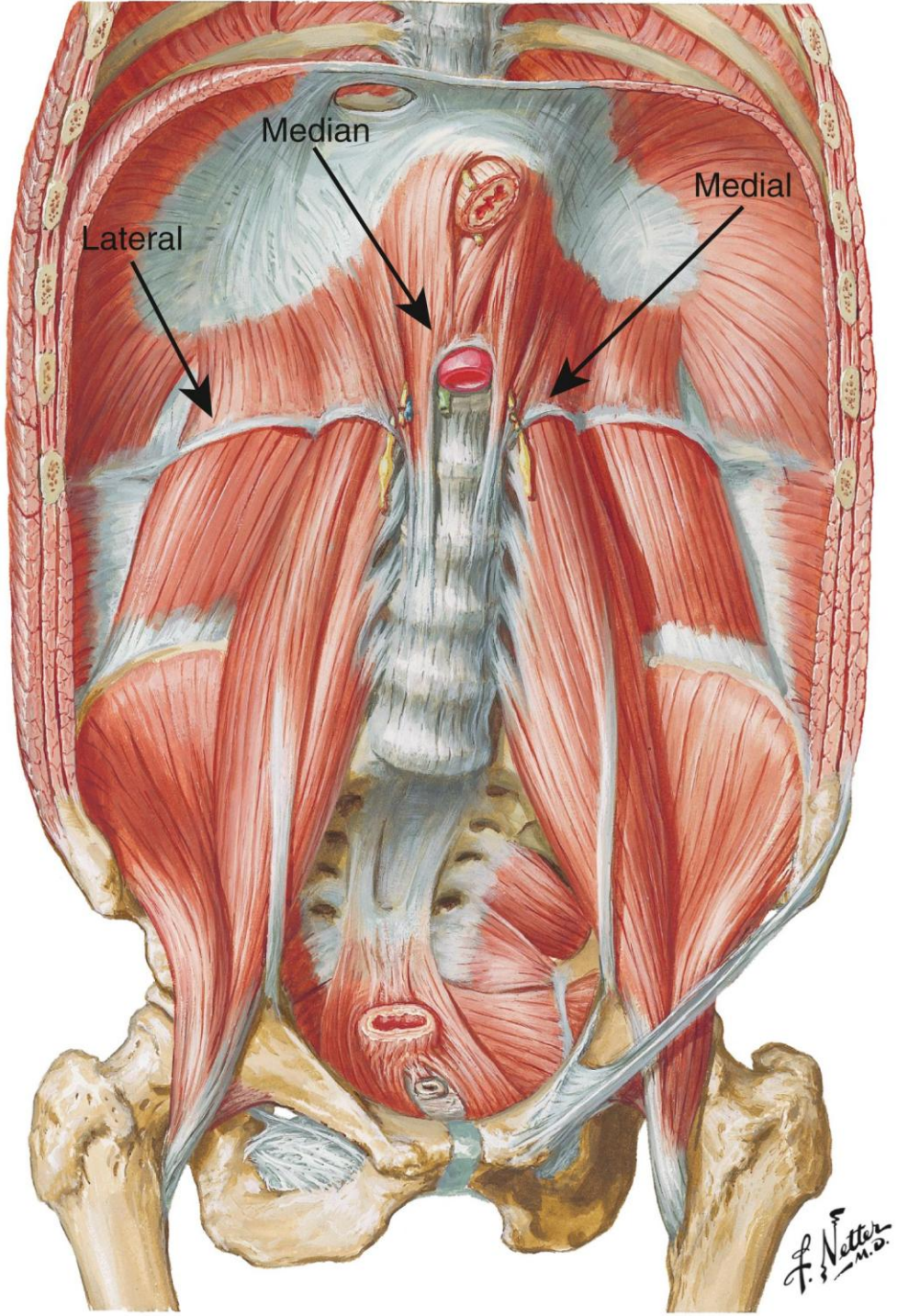
REVIEW: Muscles of the Posterior Abdominal Wall

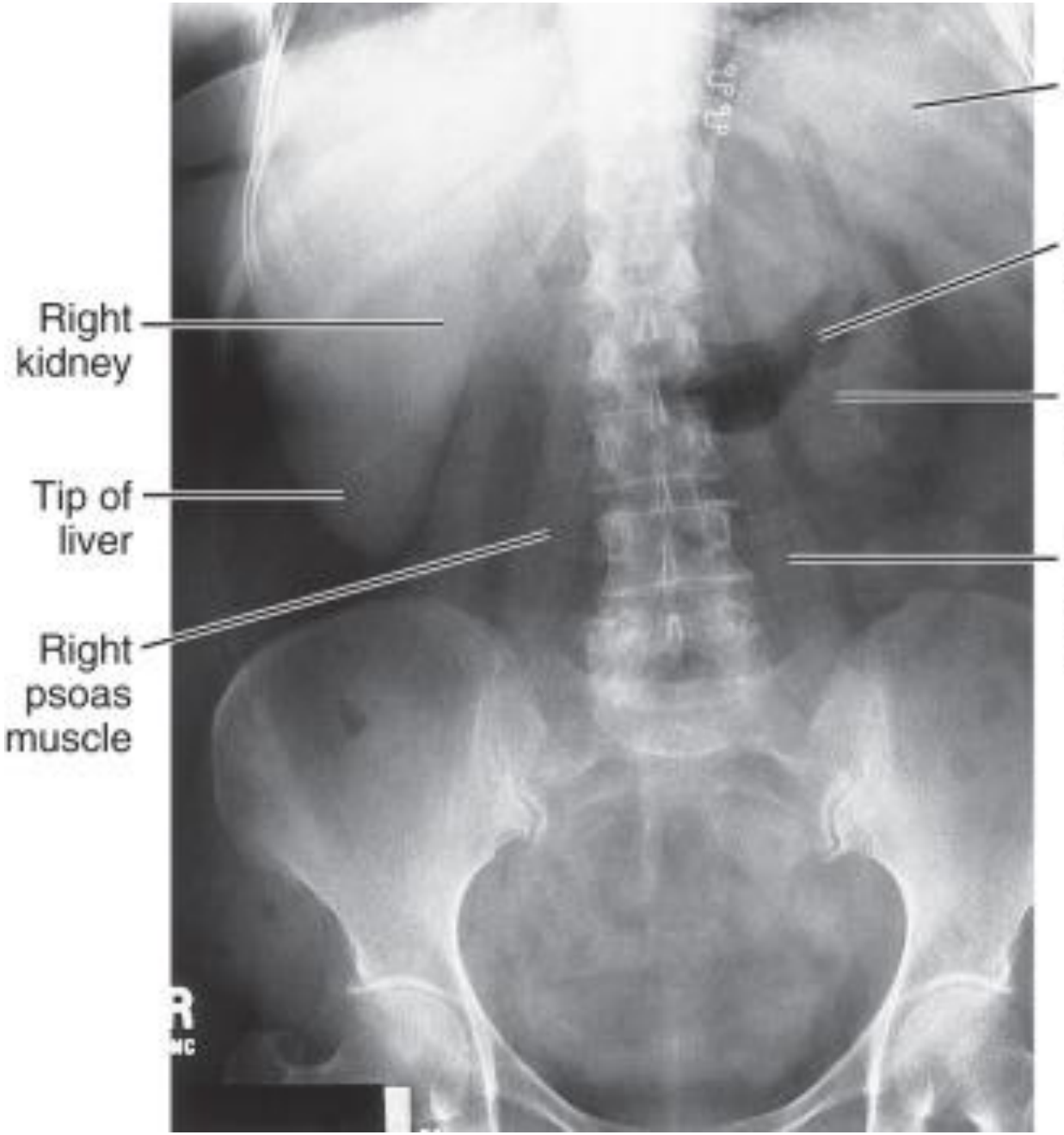
there is a muscular wall that closes the abdominal wall posteriorly



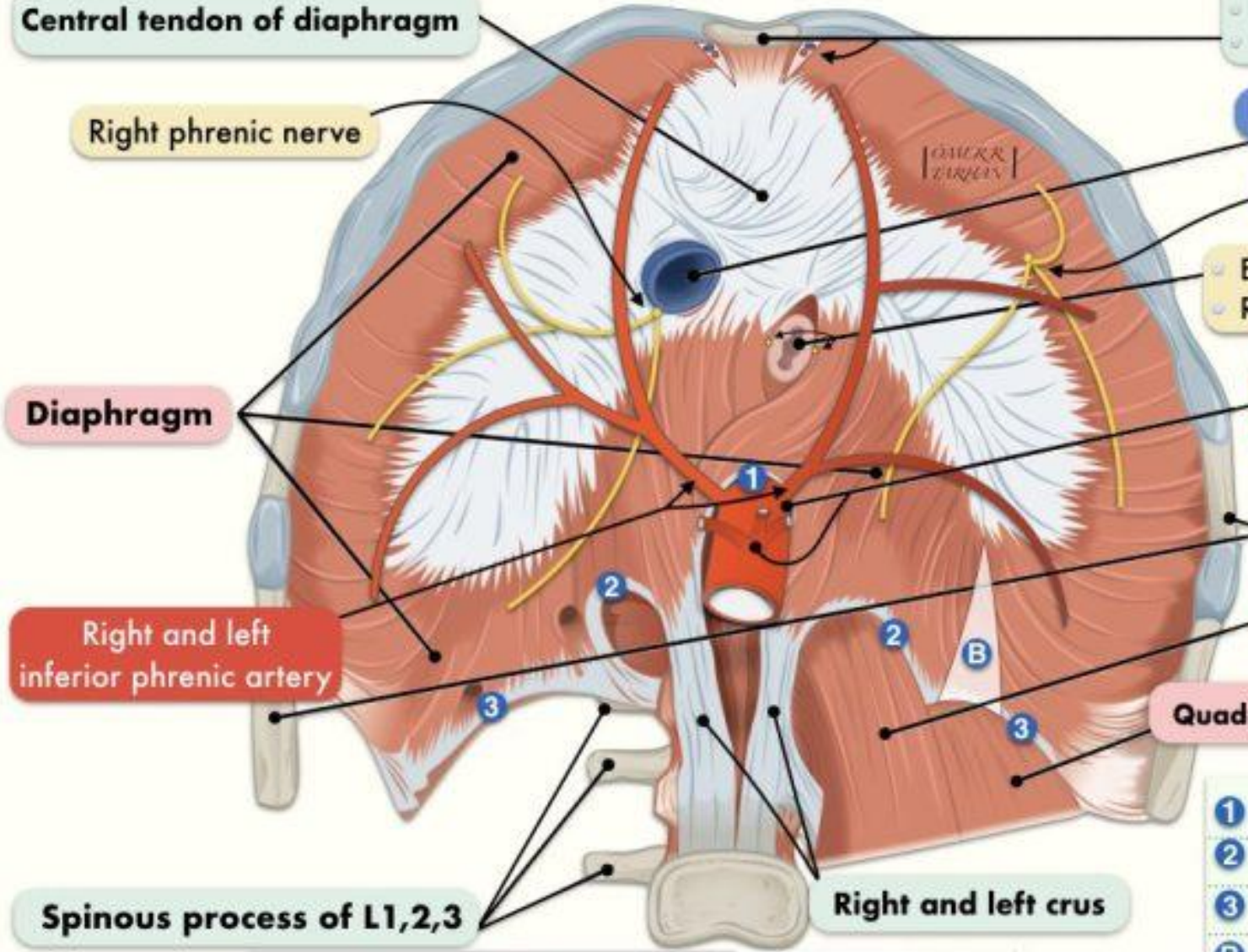


Muscle	Origin	Insertion
Psoas major	Bodies, intervertebral discs and transverse processes of T12–L5 vertebrae	Lesser trochanter of femur (through iliopsoas tendon)
Psoas minor	Bodies of T12 and L1 vertebrae and intervening disc	Pectineal line and iliopubic eminence through iliac fascia
Iliacus	Iliac fossa, iliac crest and ala of sacrum	Lesser trochanter of femur (with psoas major)
Quadratus lumborum	Iliolumbar ligament and posterior part of iliac crest	Lower border of 12th rib and transverse processes of L1–L4


Median
Medial
Lateral
F. Netter M.D.


Spleen
Gas in stomach
Right kidney
Left kidney
Tip of liver
Left psoas muscle
Right psoas muscle

Anatomy of the Diaphragm (Bottom view)



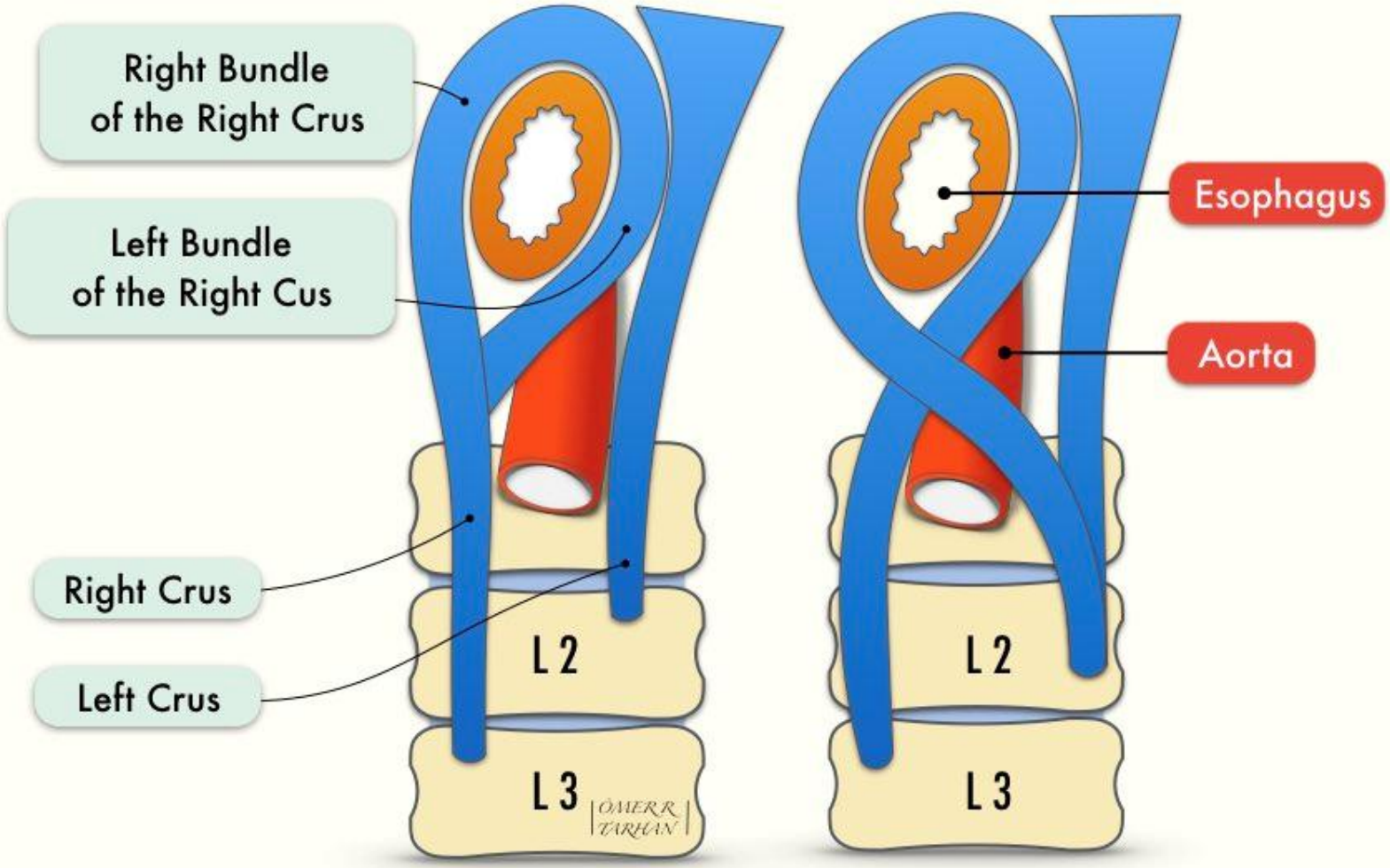
- Sternum
- Foramen of Morgagni (Sternocostal triangle)
  - Superior epigastric artery
  - Superor epigastric vein
- Central tendon of diaphragm
- Right phrenic nerve
- Inferior vena cava
- Left phrenic nerve.
- Esophagus
- Right and left vagus nerve
- Diaphragm
- Aorta
- Celiac trunk
- 12th rib
- Right and left inferior phrenic artery
- Psoas major muscle
- Quadratus lumborum muscle
- Spinous process of L1,2,3
- Right and left crus

1. Median arcuate ligament
2. Medial arcuate ligament
3. Lateral arcuate ligament
B. Foamen of Bochdalek (Lumbocostal triangle)

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Variations of Diaphragmatic Crura

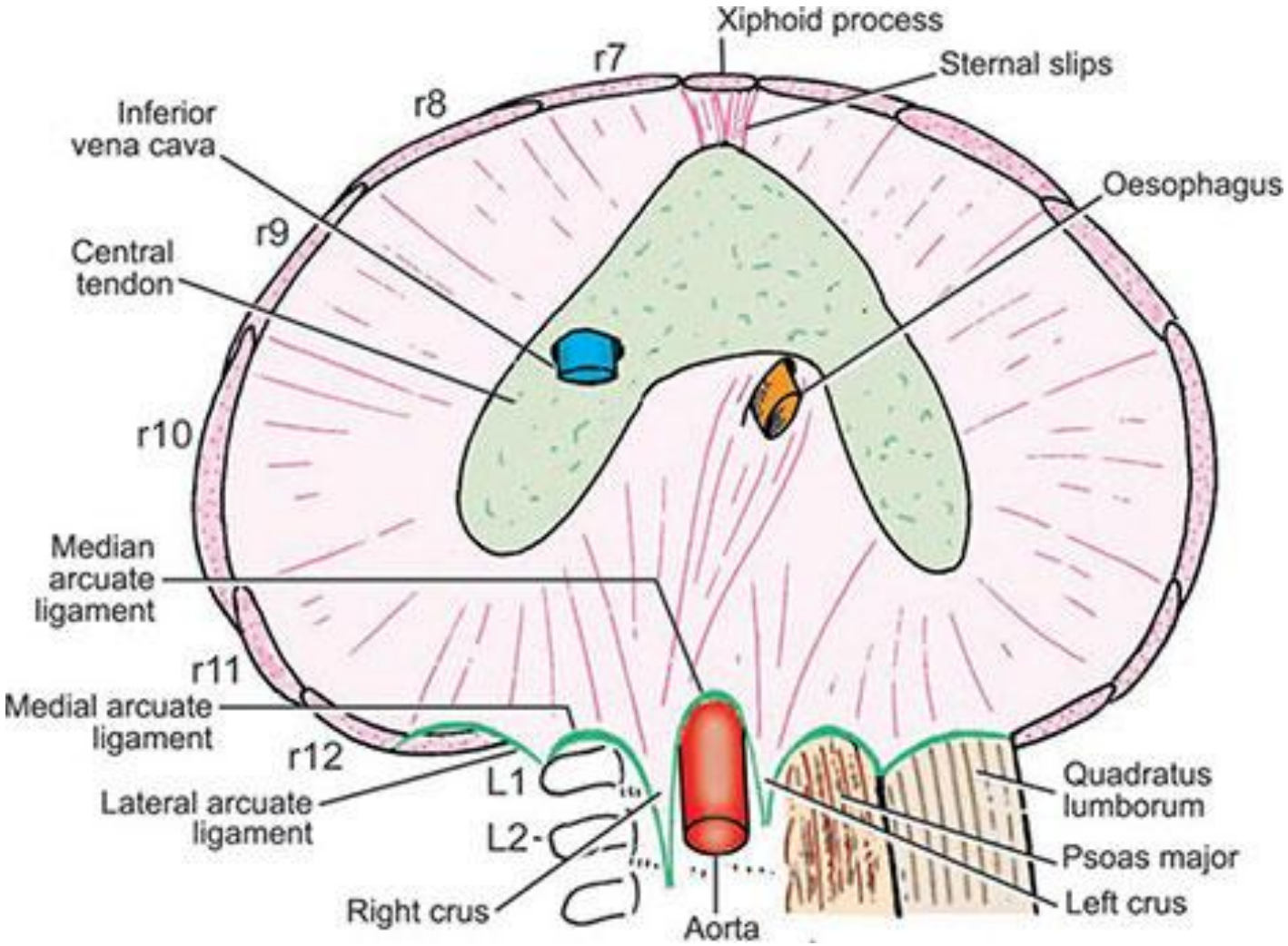


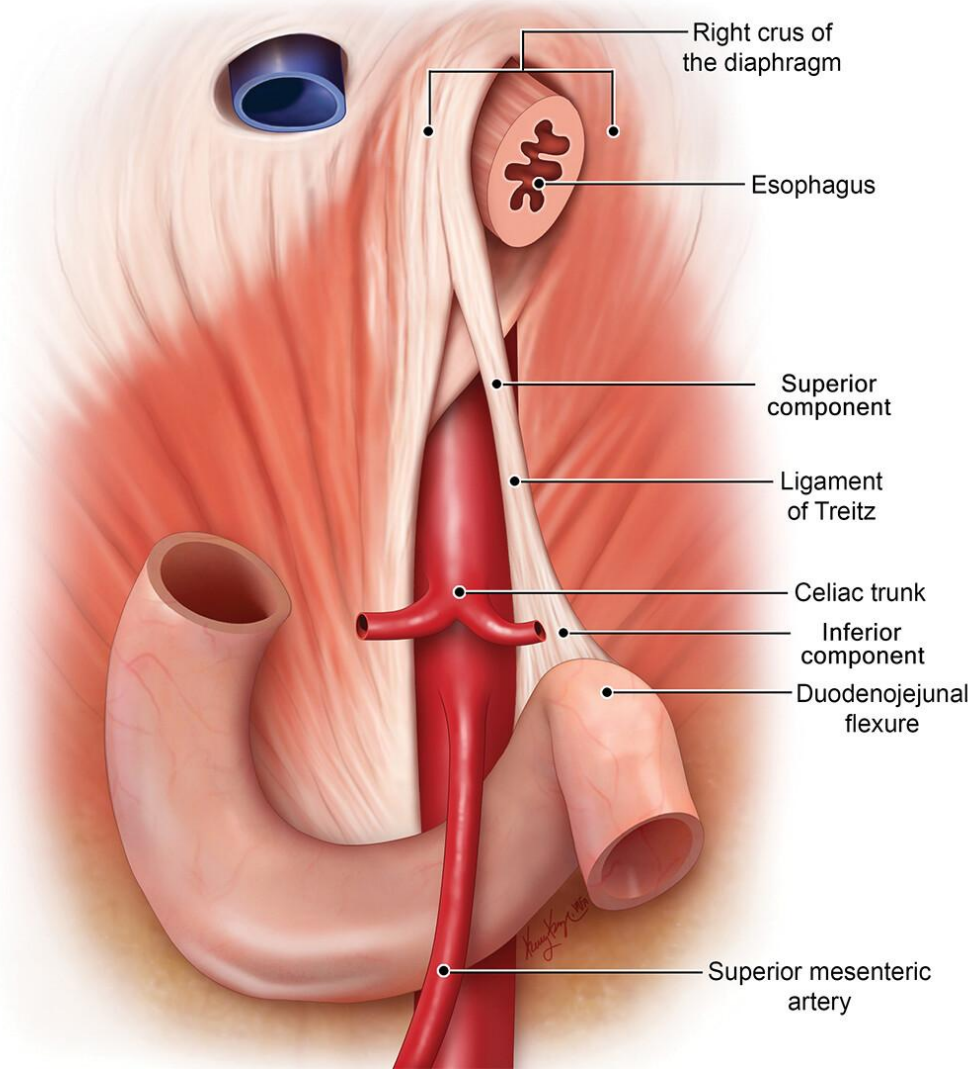
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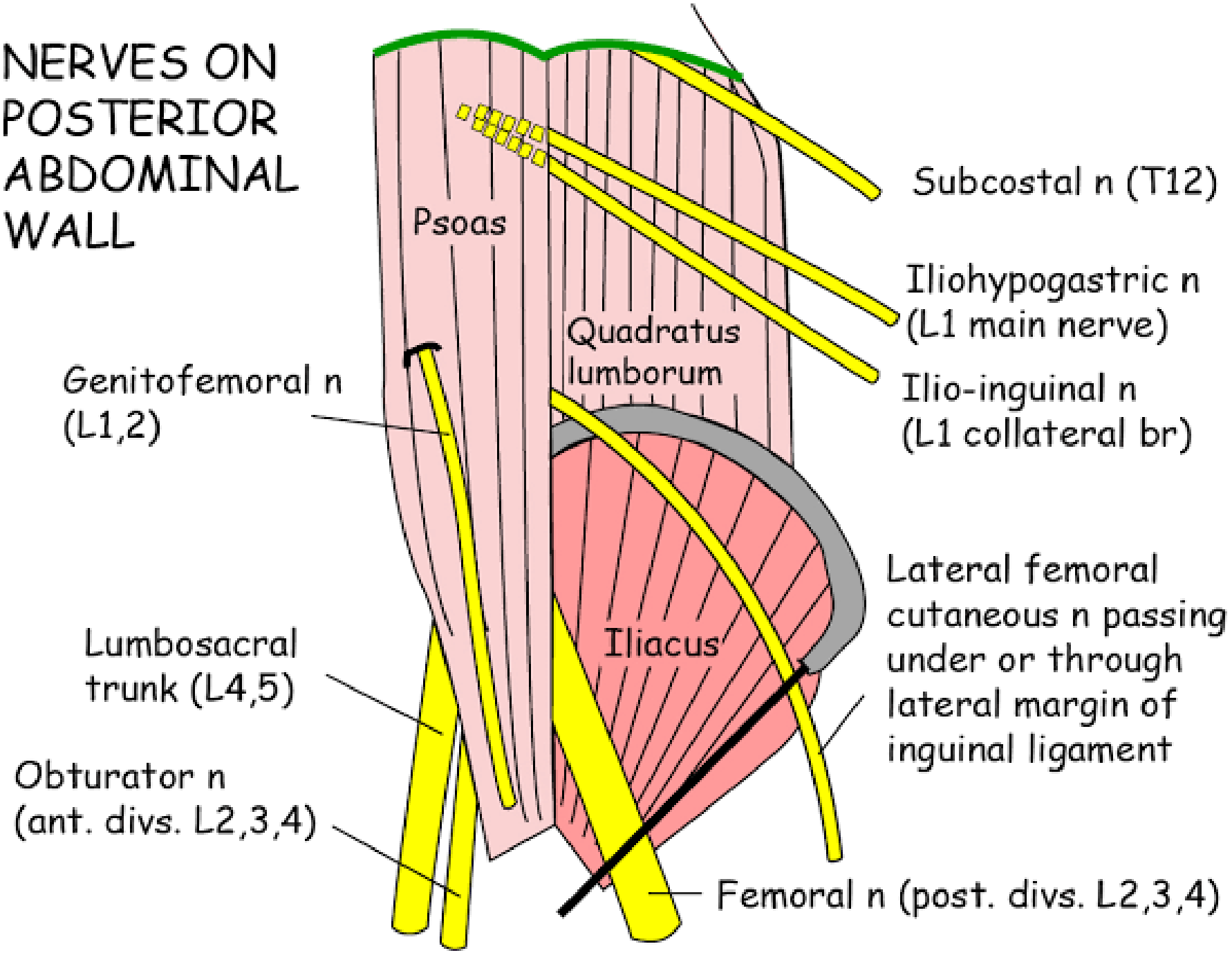
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Xiphoid process
r7
Sternal slips
r8
Inferior vena cava
Oesophagus
r9
Central tendon
r10
Median arcuate ligament
r11
Medial arcuate ligament
r12
Lateral arcuate ligament
L1
L2
Quadratus lumborum
Psoas major
Left crus
Right crus
Aorta


Right crus of the diaphragm
Esophagus
Superior component
Ligament of Treitz
Celiac trunk
Inferior component
Duodenojejunal flexure
Superior mesenteric artery

NERVES ON POSTERIOR ABDOMINAL WALL



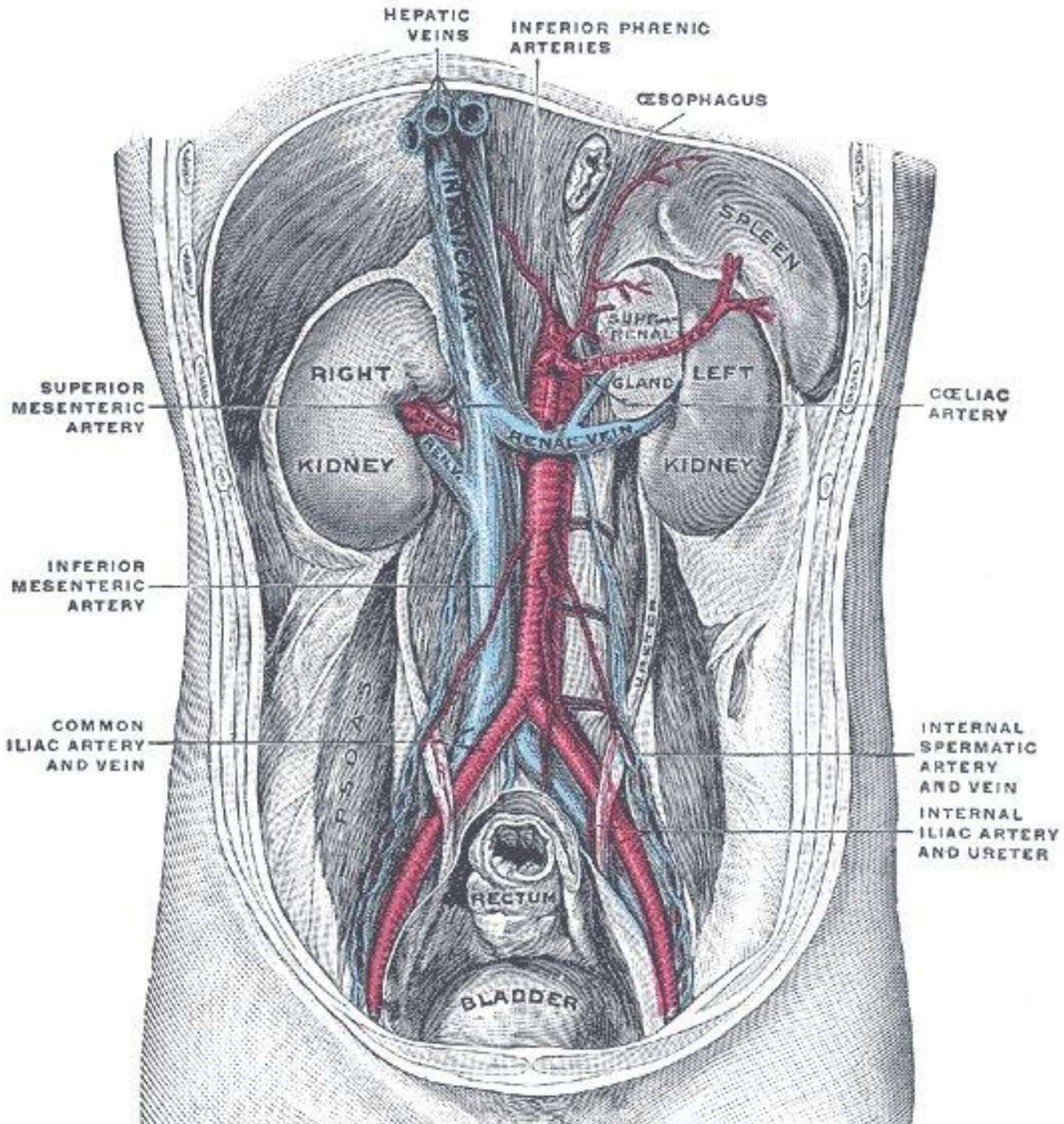
Blood Vessels

Arteries:

- Abdominal aorta.
- Lumbar arteries.
- Common iliac arteries.
- Median sacral artery.
- Renal arteries.

Veins:

- Inferior vena cava.
- Lumbar veins.
- Common iliac veins.
- Renal veins.



KIDNEYS:

Renal Functions



Regulation of red blood cell production



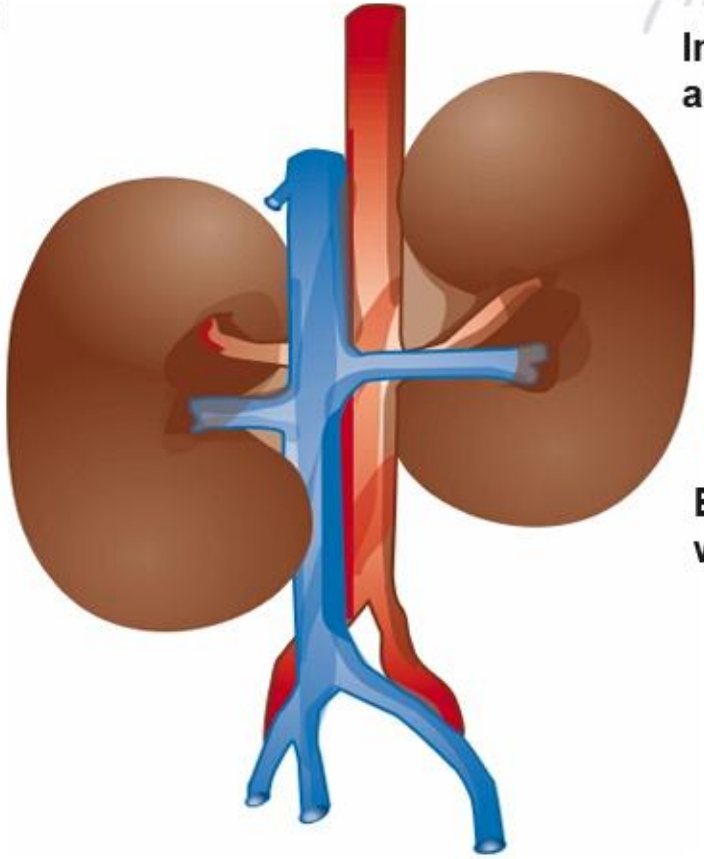
Regulation of blood pressure



Influence on blood pH acid-base-metabolism

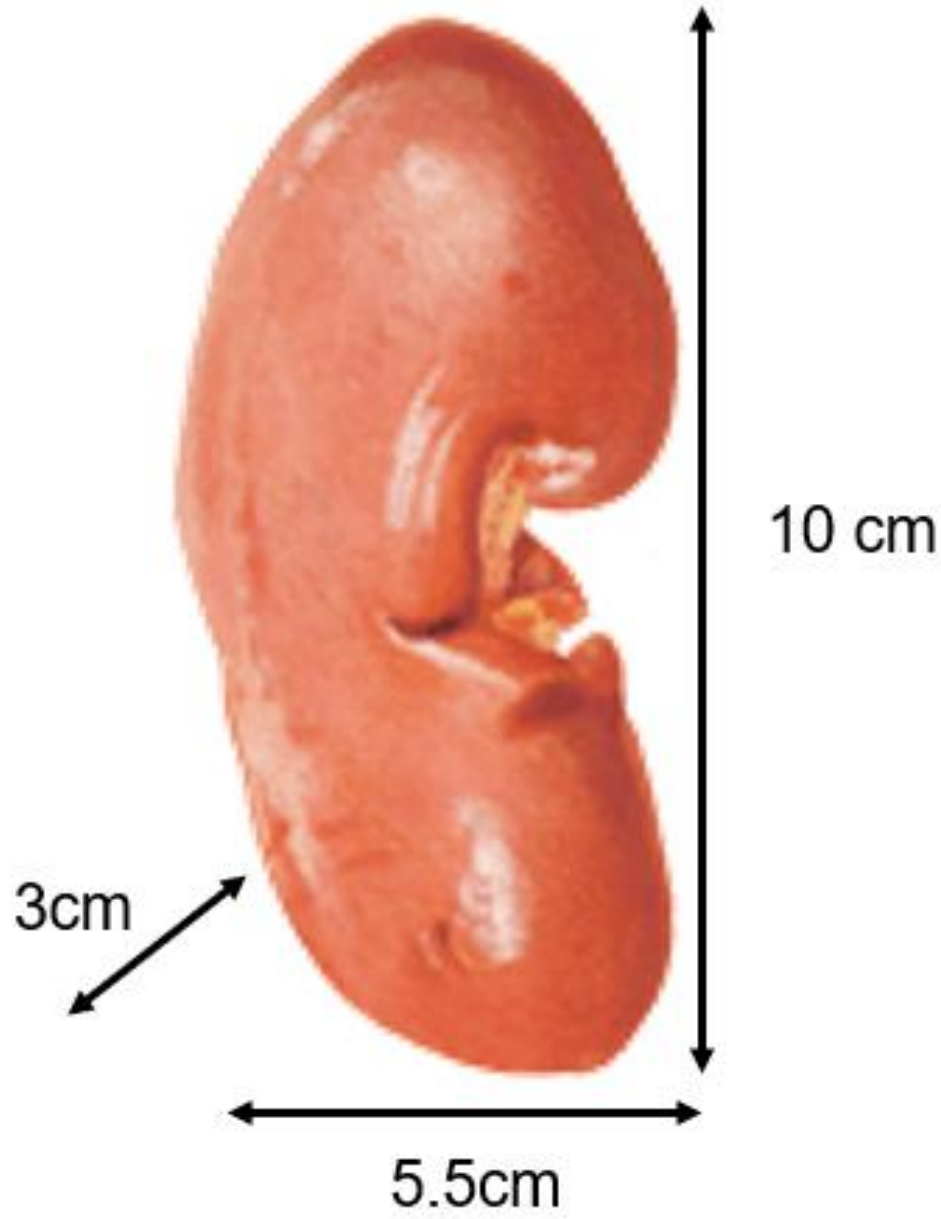


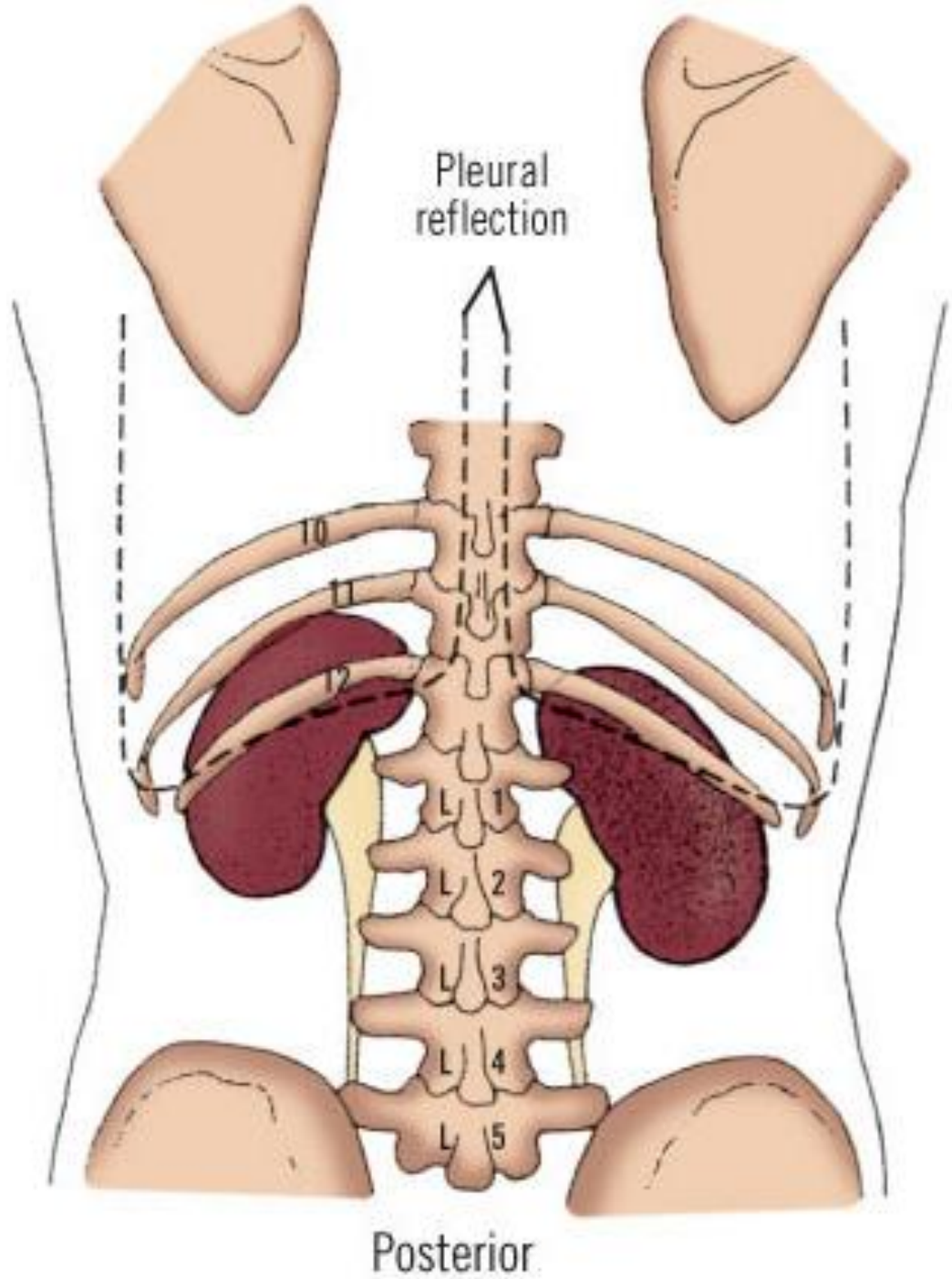
Regulation of bone-mineral-metabolism

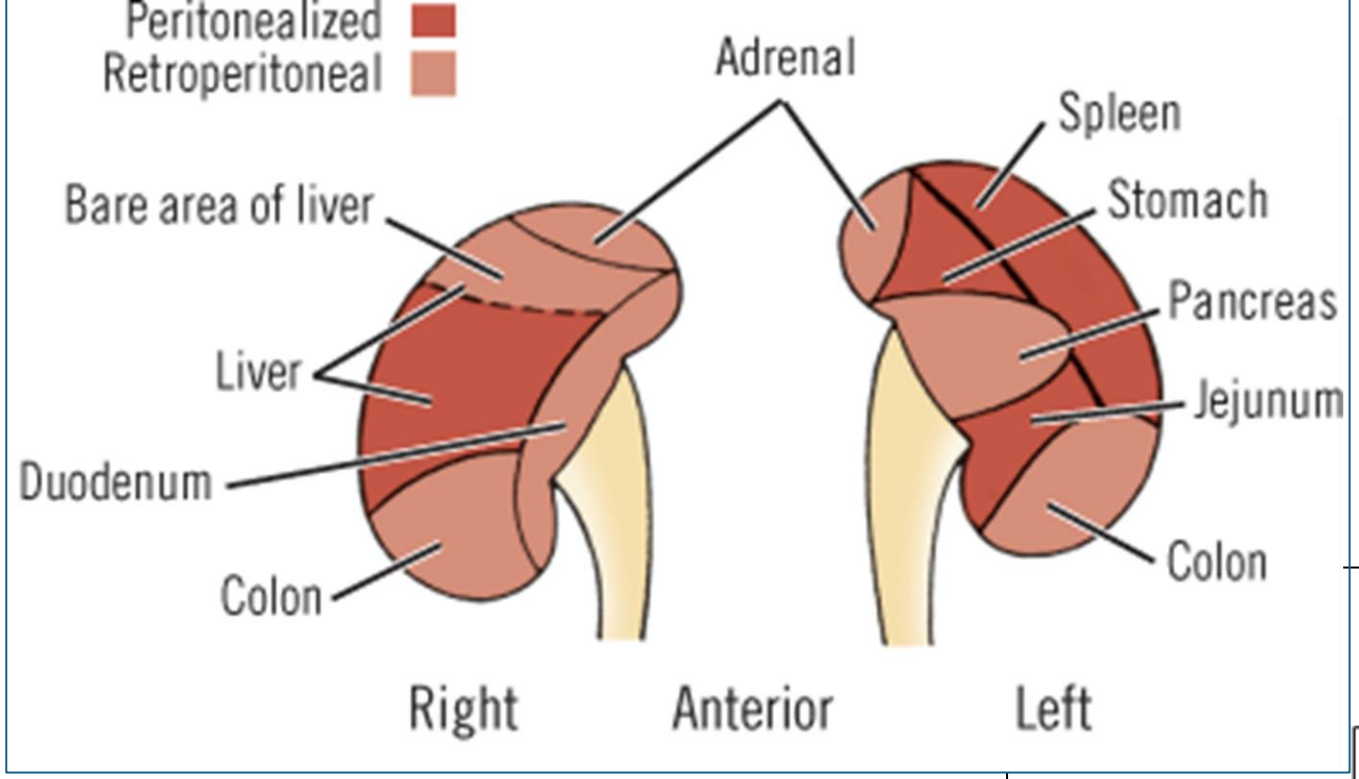


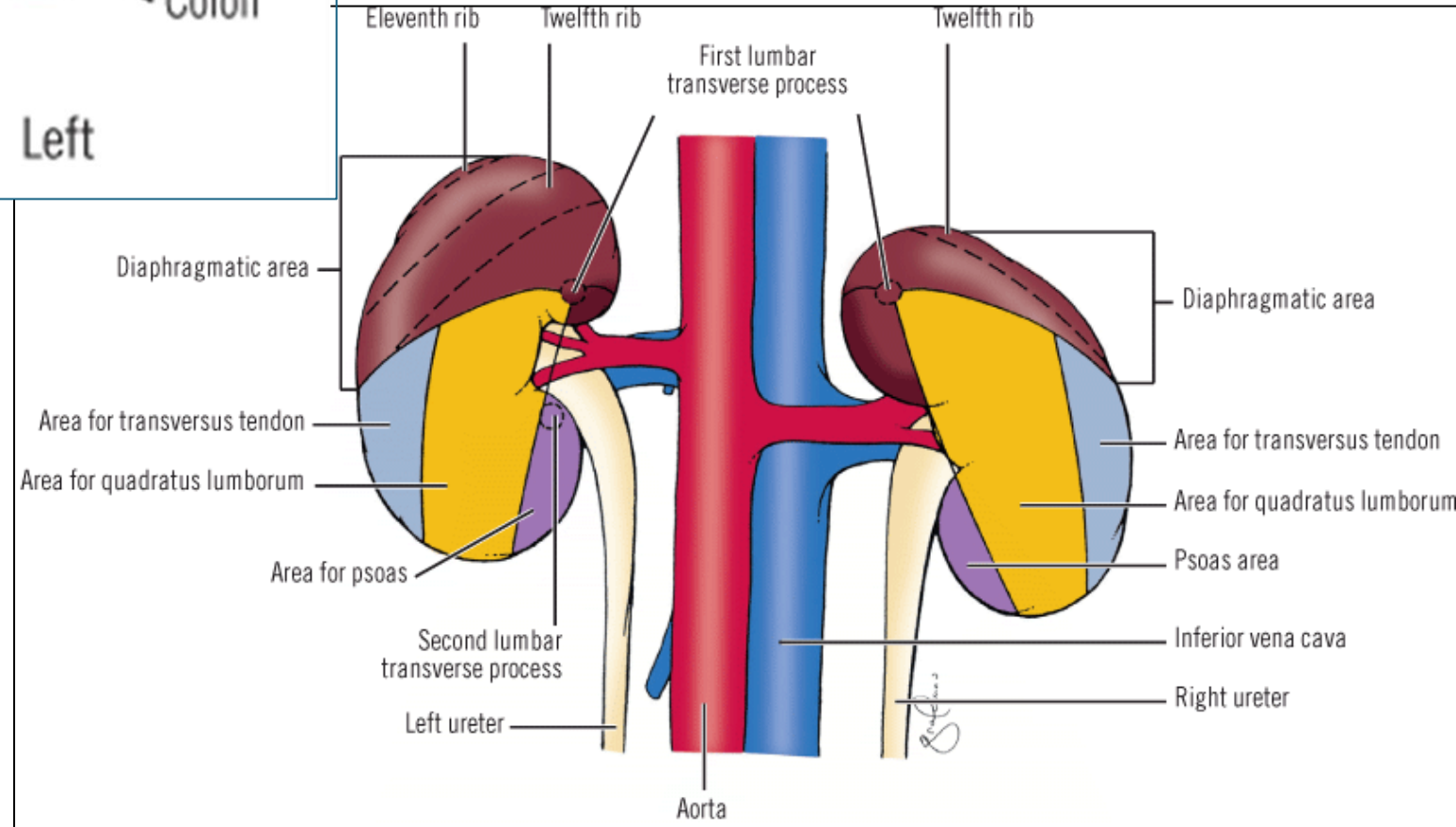


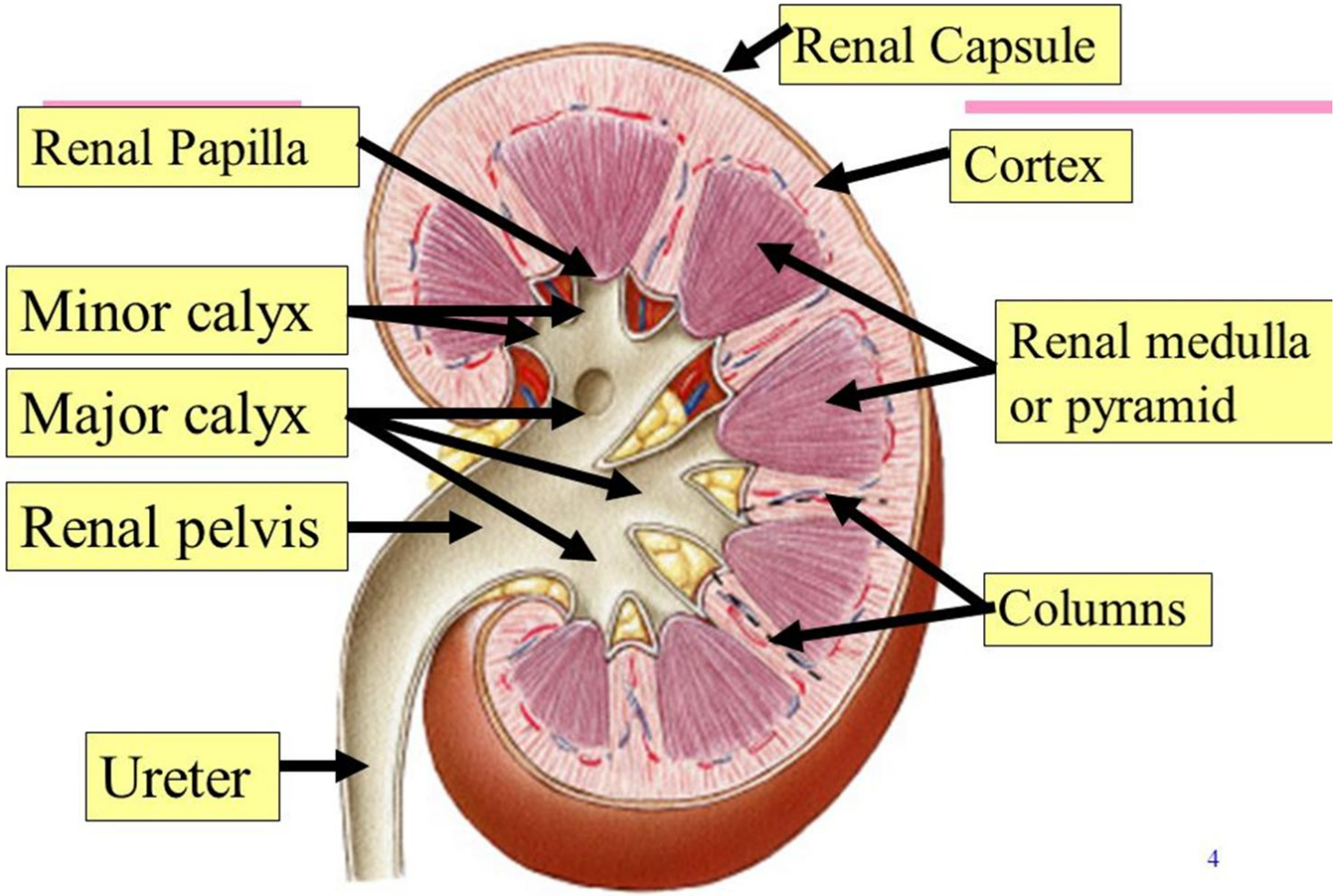
Excretion of metabolic waste products and water

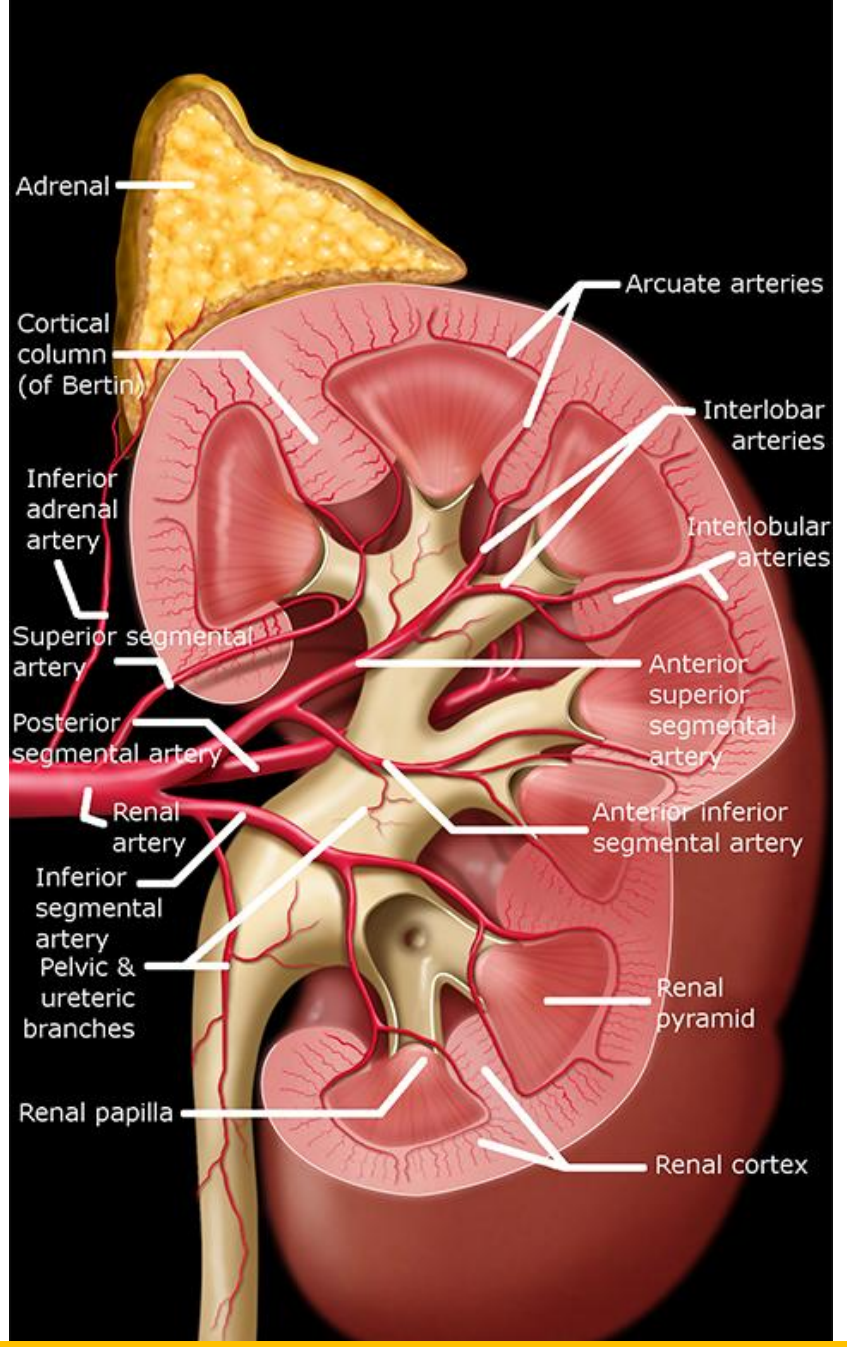

10 cm
3cm
5.5cm


Pleural reflection
10
11
12
L 1
L 2
L 3
L 4
L 5
Posterior

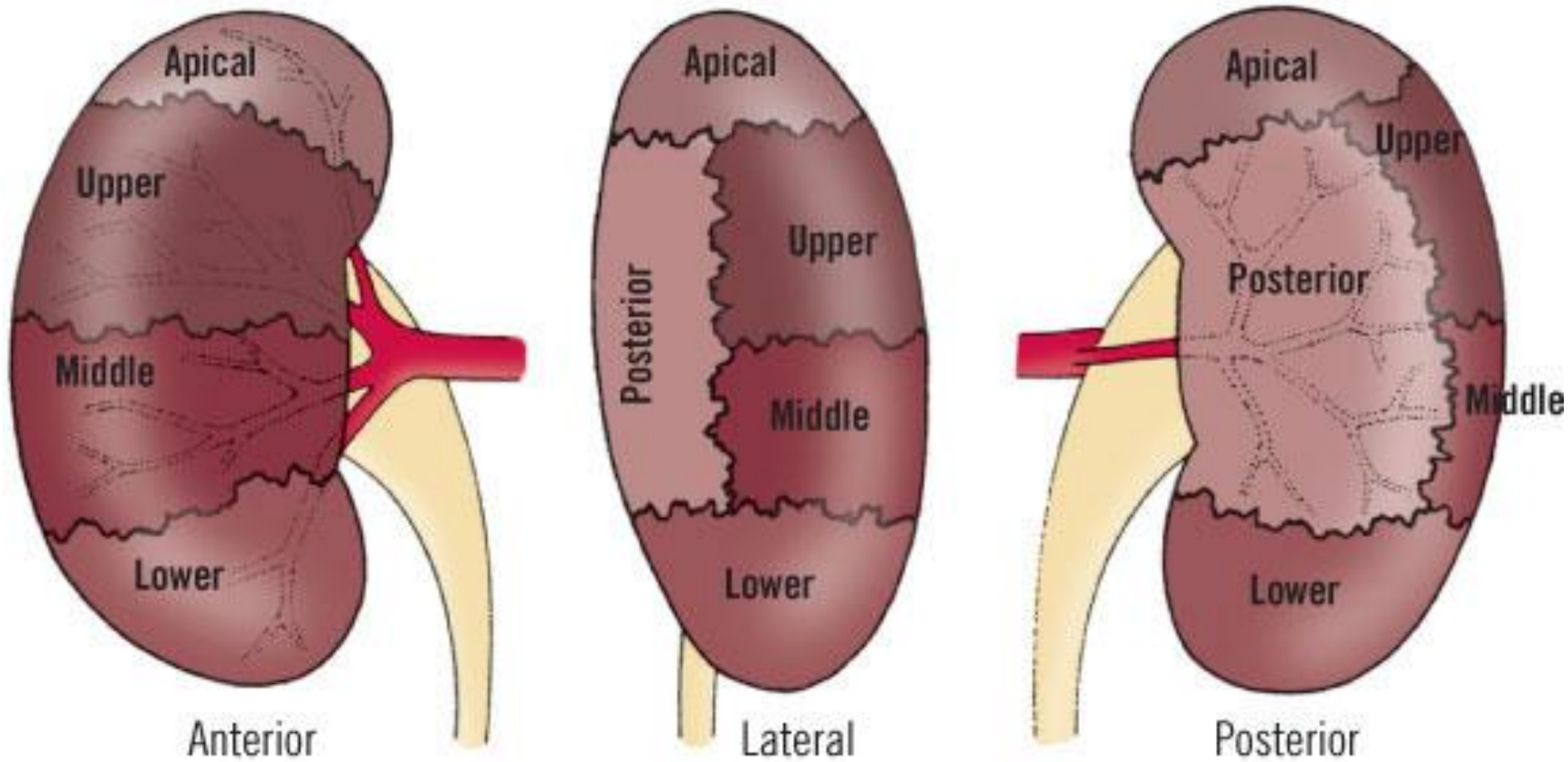

Peritonealized
Retroperitoneal
Adrenal
Spleen
Bare area of liver
Stomach
Pancreas
Liver
Jejunum
Duodenum
Colon
Colon
Right
Anterior
Left


Eleventh rib
Twelfth rib
First lumbar transverse process
Twelfth rib
Diaphragmatic area
Diaphragmatic area
Area for transversus tendon
Area for transversus tendon
Area for quadratus lumborum
Area for quadratus lumborum
Psoas area
Area for psoas
Second lumbar transverse process
Inferior vena cava
Right ureter
Left ureter
Aorta



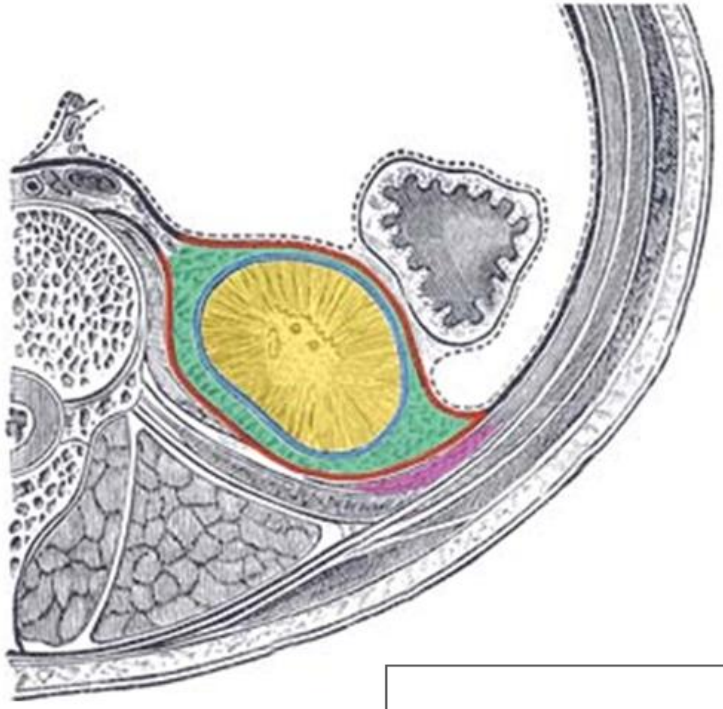


Intrarenal Subdivision of the renal artery



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Renal coverings



Kidney

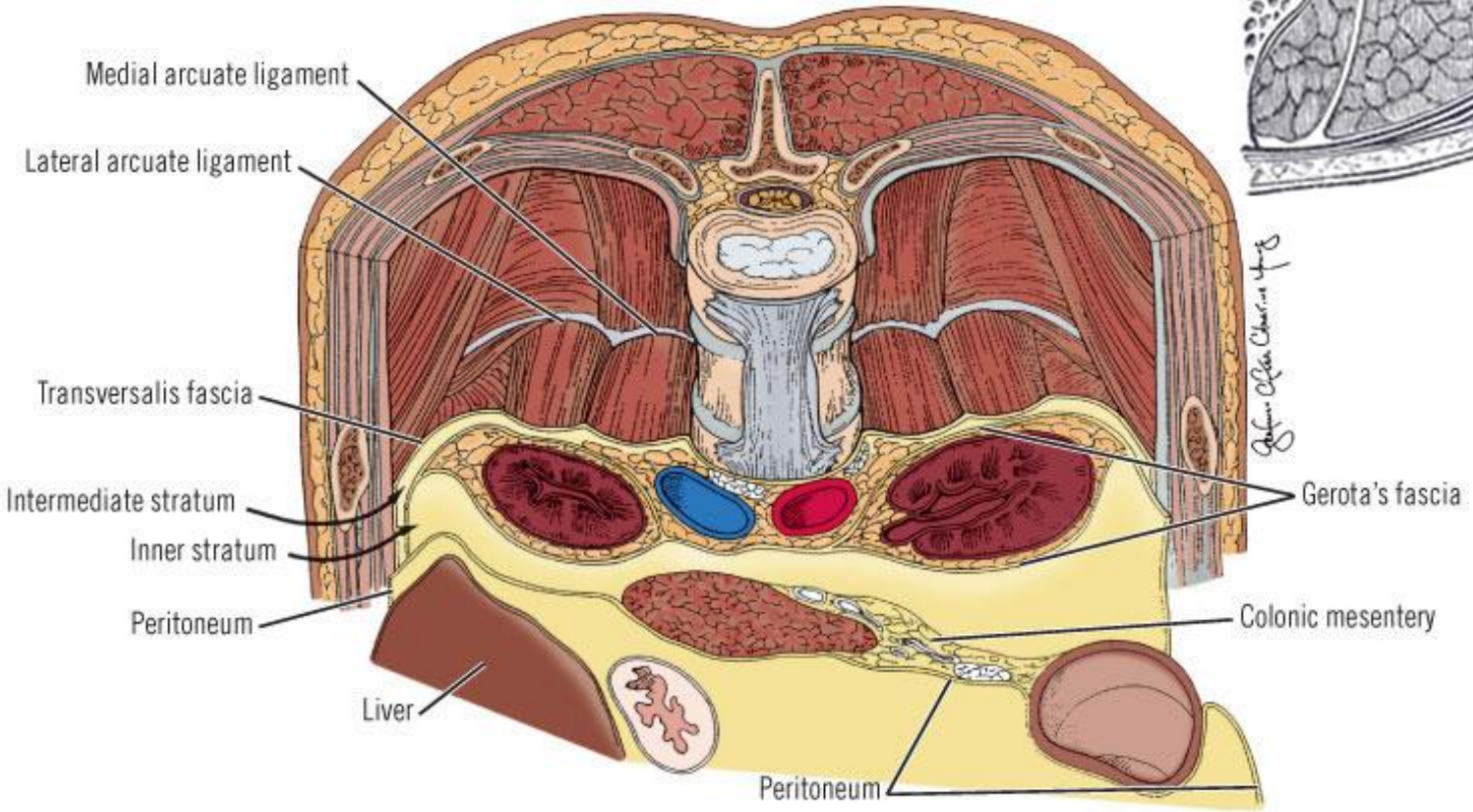
Renal capsule

Perirenal fat

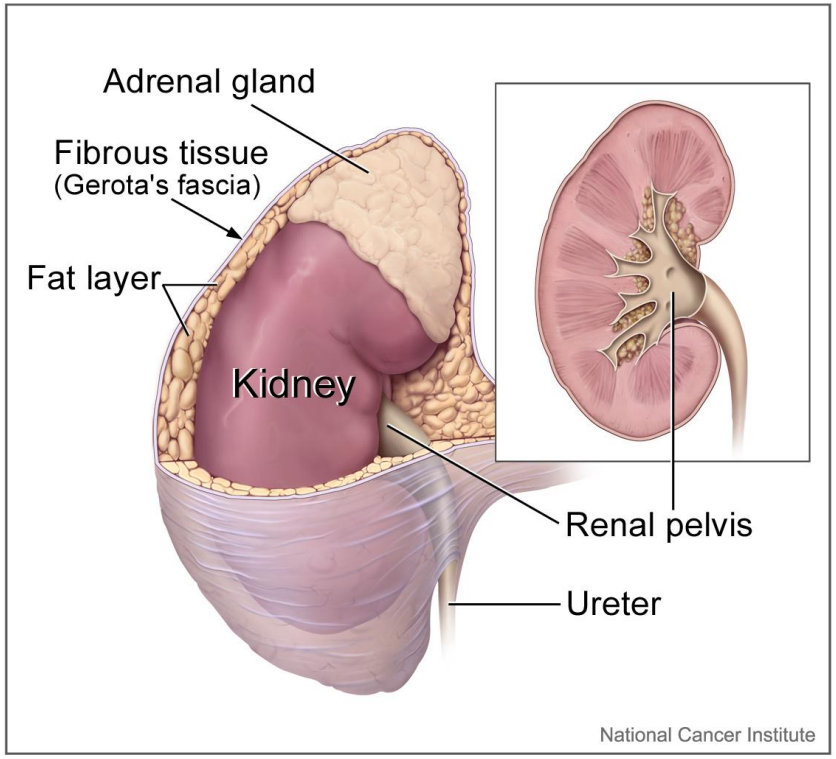
Renal fascia

= Gerota's Fascia

Pararenal fat

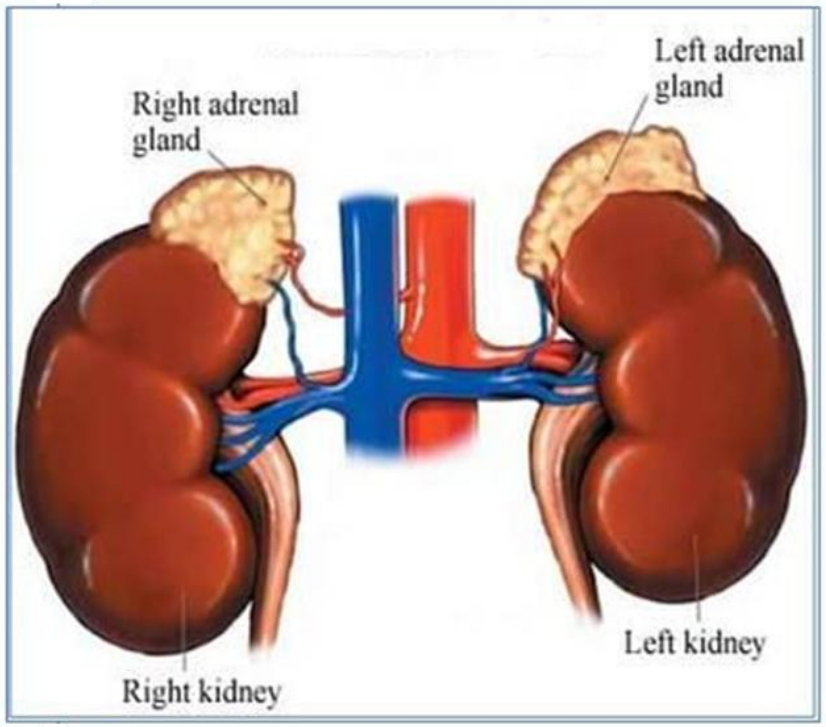


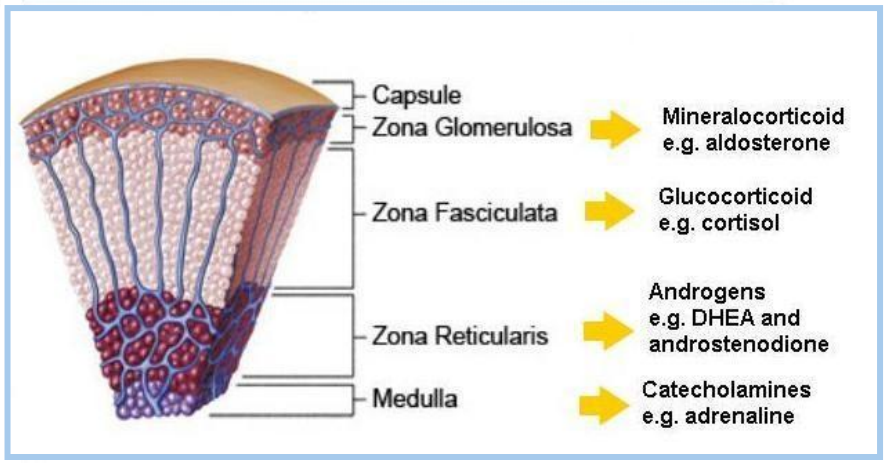
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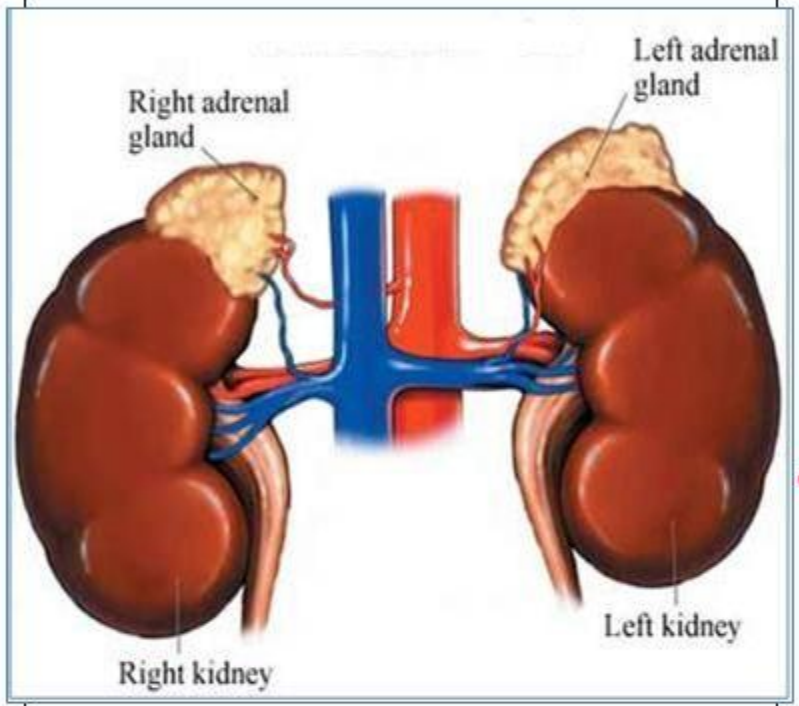
Suprarenal Glands

- The **suprarenal (adrenal) gland** is a component of the **hypothalamic-pituitary-suprarenal axis** that is responsible for coordinating **stress response** and **metabolism**.
- They are yellowish retroperitoneal organs that lie on the upper poles of the kidneys,
- just above the level of the last thoracic vertebra (T12).
- They are surrounded by renal fascia (but are separated from the kidneys by the perirenal fat).
- Each gland has an outer yellow cortex and an inner dark brown medulla.

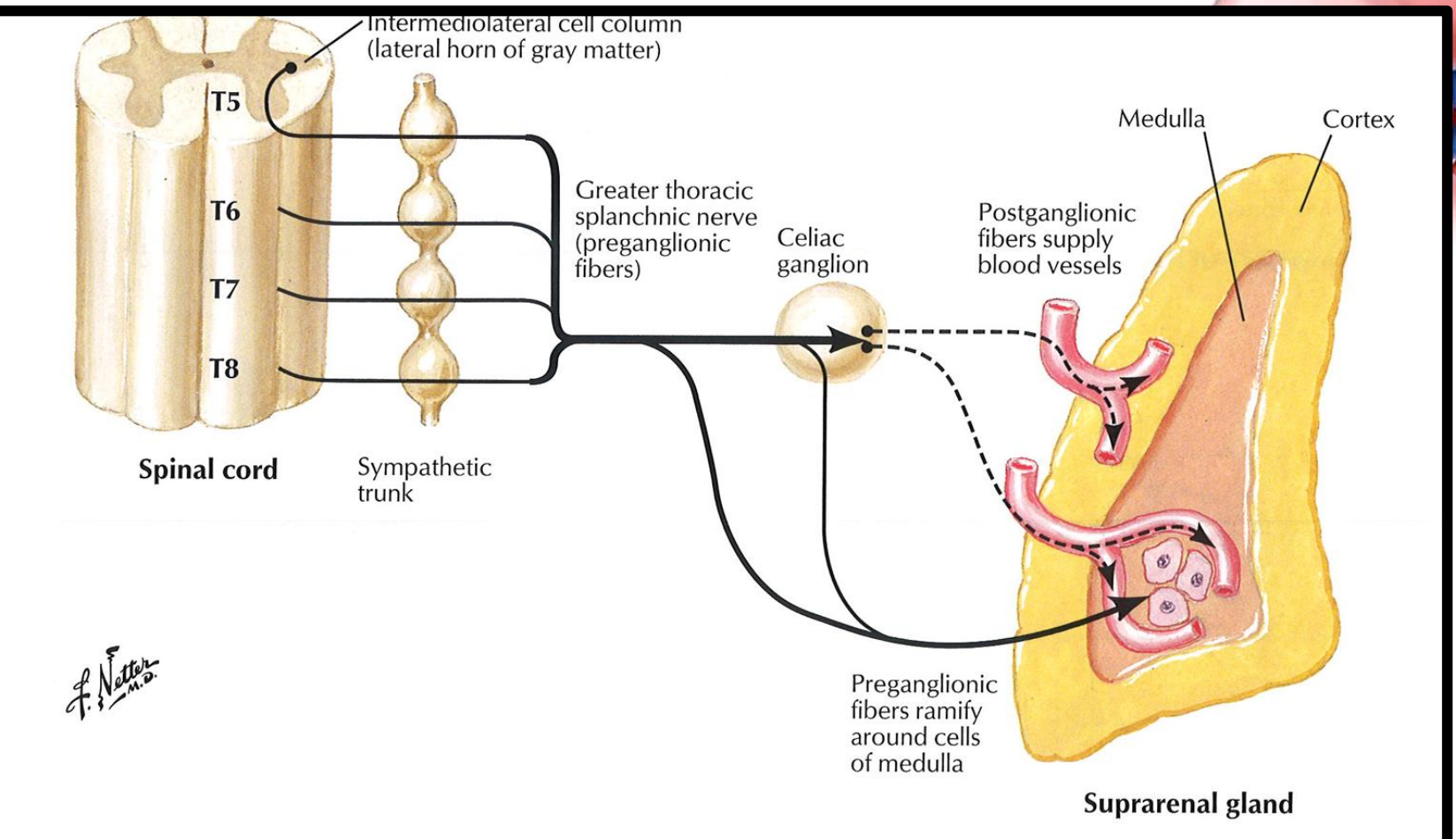


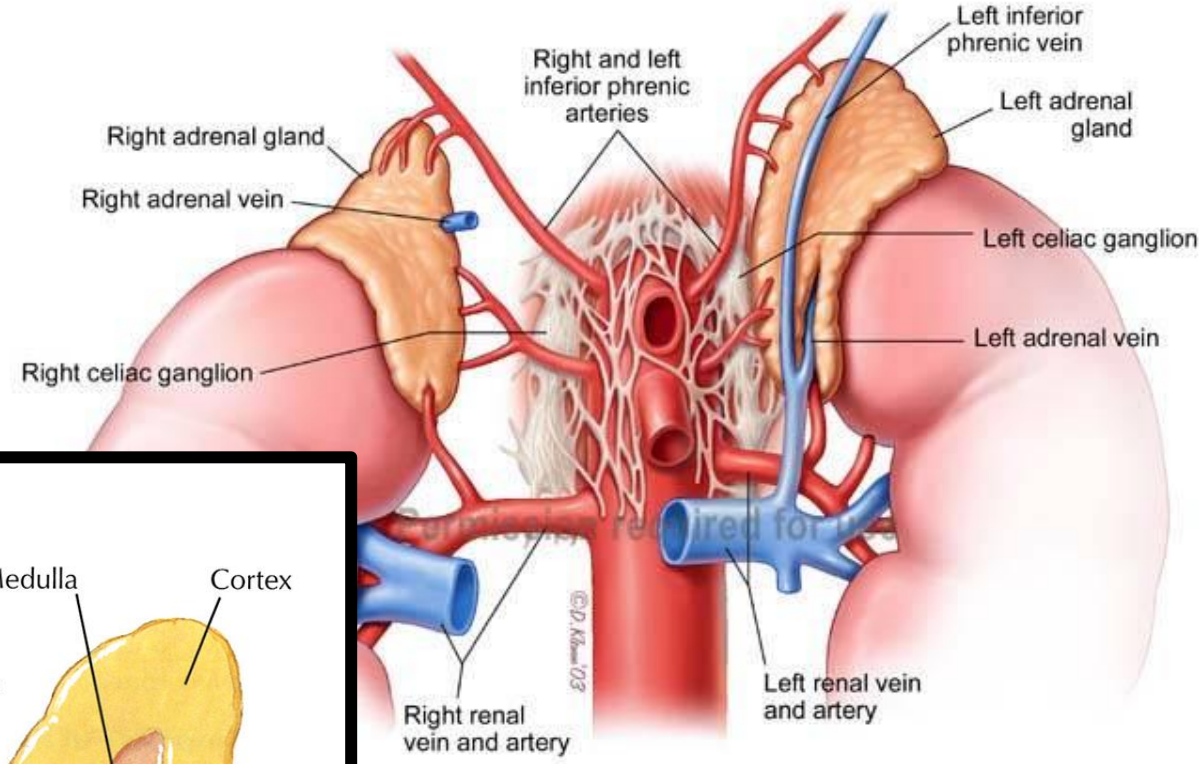


- The **right gland** is pyramid shaped and caps the upper pole of the right kidney.
- ***Relations:***
  - **Anterior:** right lobe of the liver and inferior vena cava.
  - **Posterior:** diaphragm.



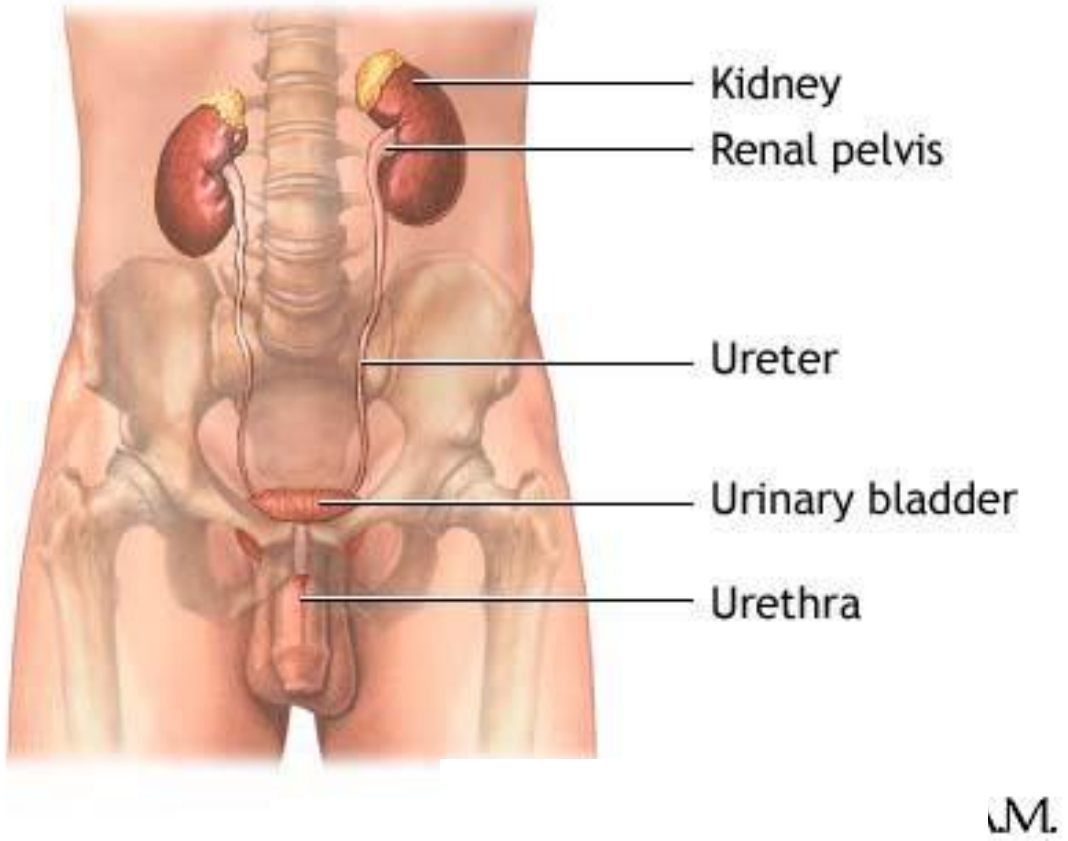
- The **left gland** is crescent in shape and extends along the **medial border** of the left kidney from the upper pole to the hilus.
- ***Relations:***
  - **Anterior:** pancreas, lesser sac, and stomach
  - **Posterior:** diaphragm.


Intermediolateral cell column
(lateral horn of gray matter)
T5
T6
T7
T8
Spinal cord
Sympathetic trunk
Greater thoracic splanchnic nerve (preganglionic fibers)
Celiac ganglion
Postganglionic fibers supply blood vessels
Medulla
Cortex
Preganglionic fibers ramify around cells of medulla
Suprarenal gland


Right and left inferior phrenic arteries
Left inferior phrenic vein
Left adrenal gland
Right adrenal gland
Right adrenal vein
Left celiac ganglion
Left adrenal vein
Right celiac ganglion
Right renal vein and artery
Left renal vein and artery

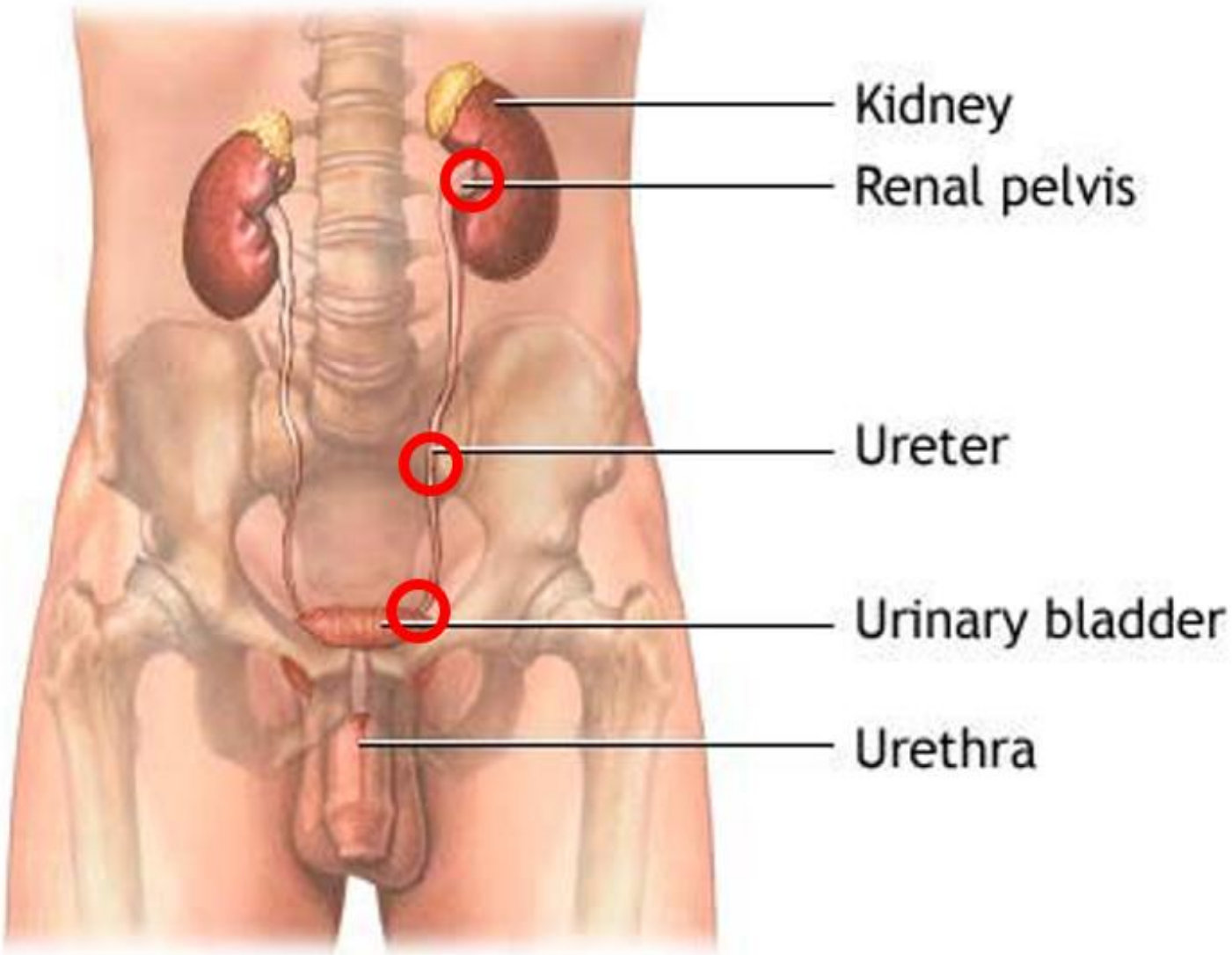
The Ureters

- Continuous with R. pelvis
- Pass behind peritoneum
- Infront of the transverse processes of lumbar vertebrae
- In the pelvic cavity enter the posterior wall of the bladder

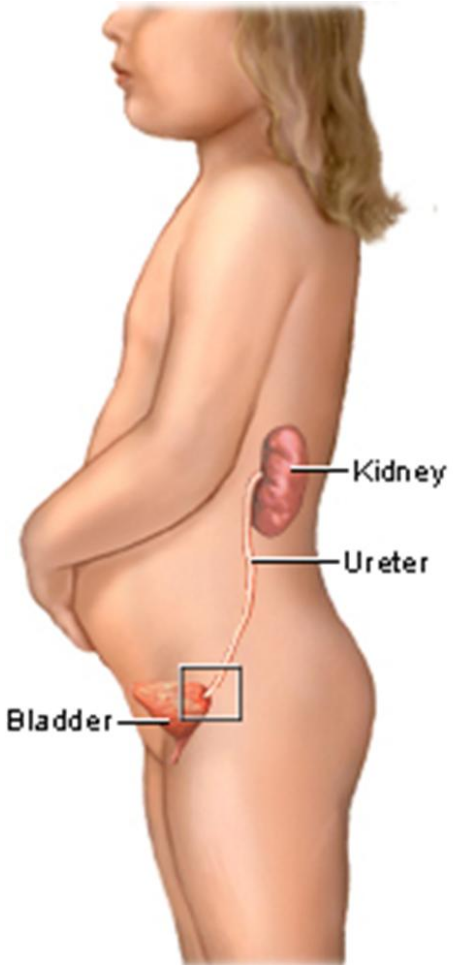


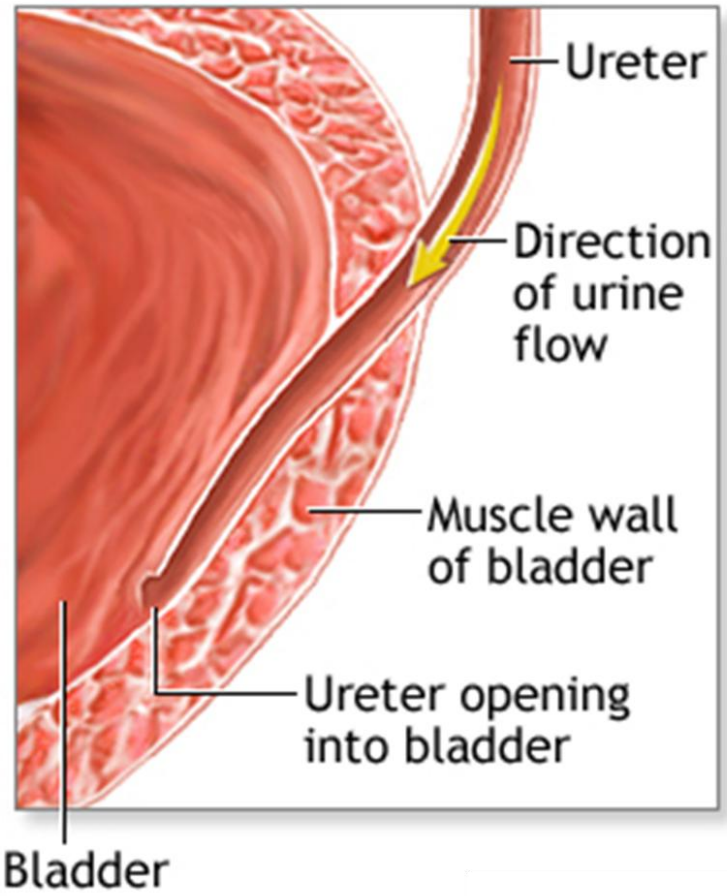
25-30 cm in length

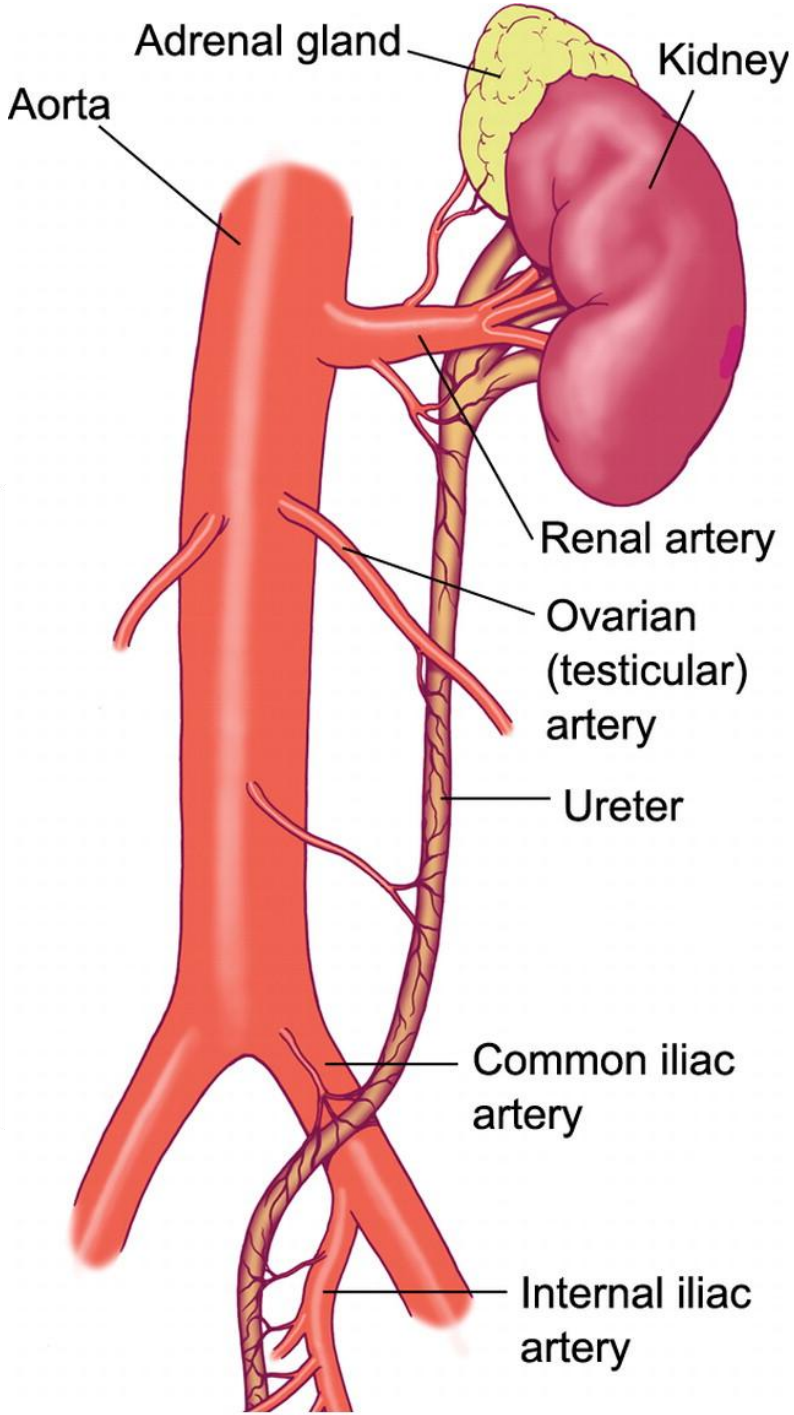
Ureteric Constrictions





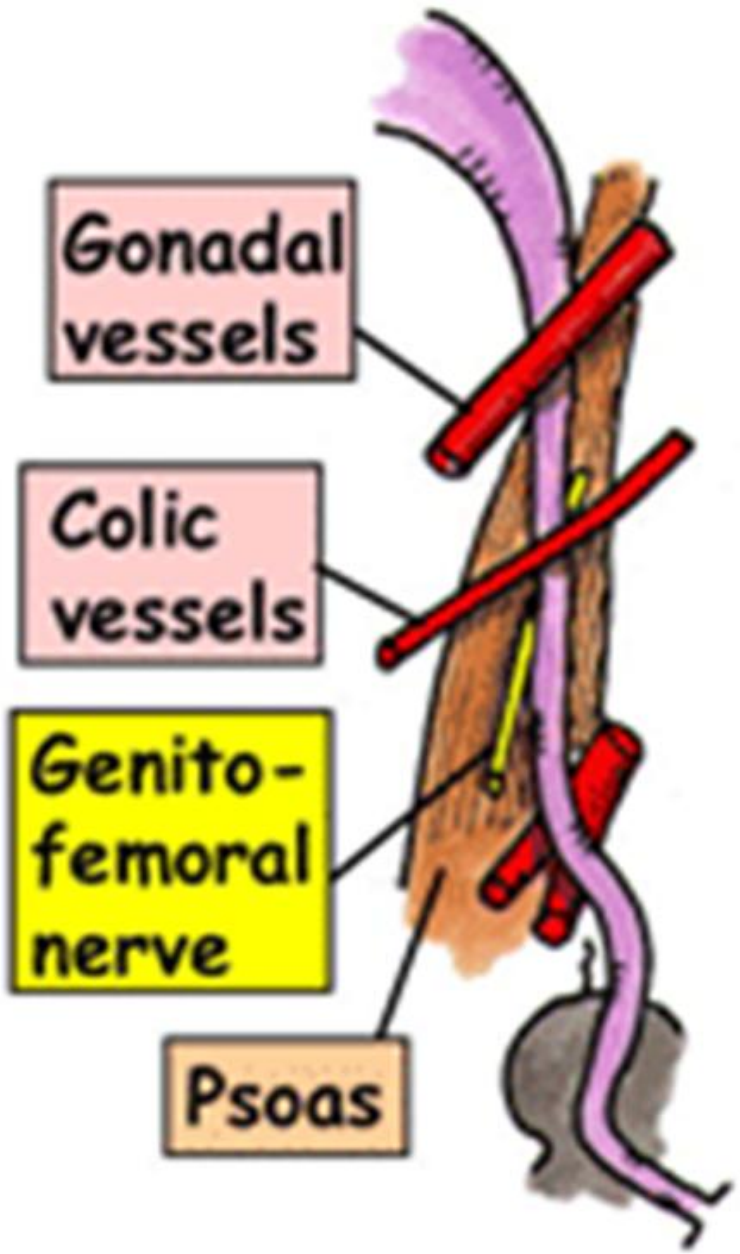




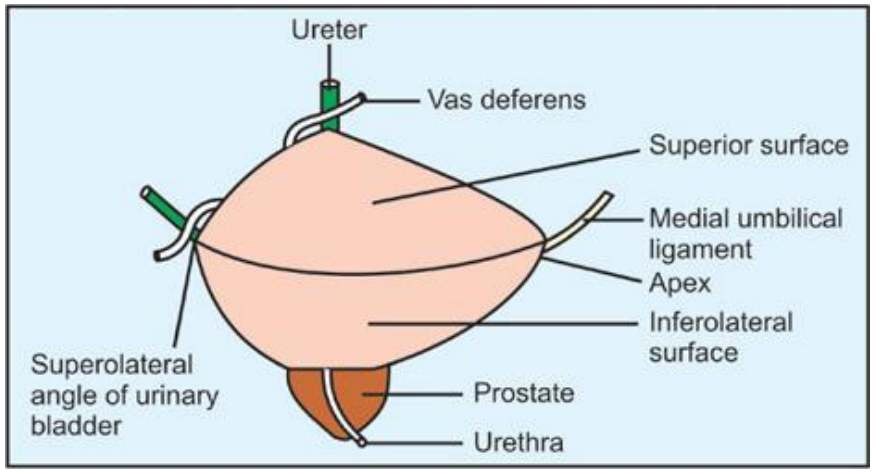


Ureteric Arteries

- Posterior: Psoas, genitofemoral nerve sacroiliac joint, common iliac artery bifurcation
- Anterior: Right- Duodenum, right gonadal artery, right colic artery, ileal mesentery, superior mesenteric artery
  Left- Left gonadal artery, left colic artery, sigmoid mesentery
- Passing under it: Vas, uterine artery
- Related to it: Lateral fornix in female



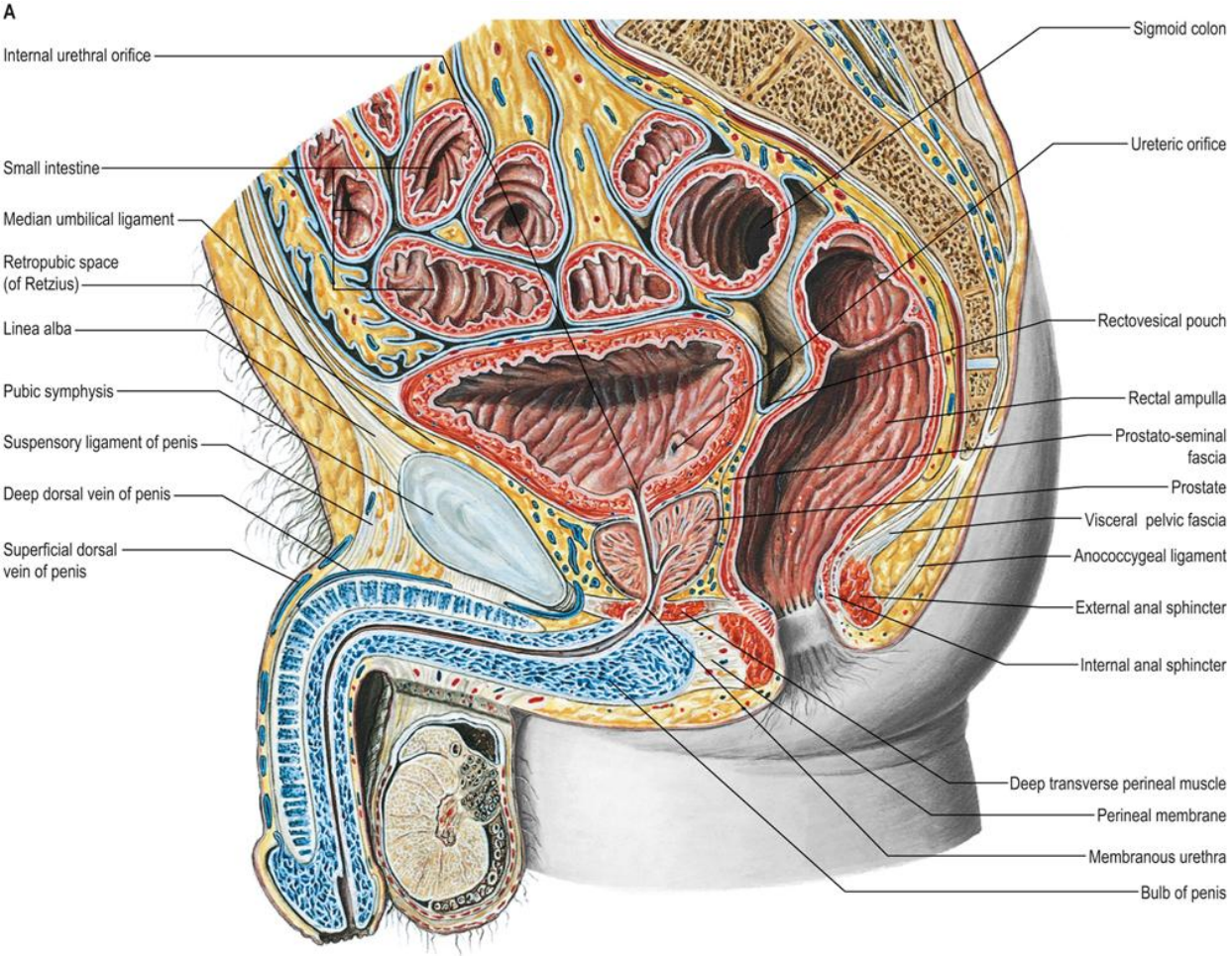
Urinary Bladder



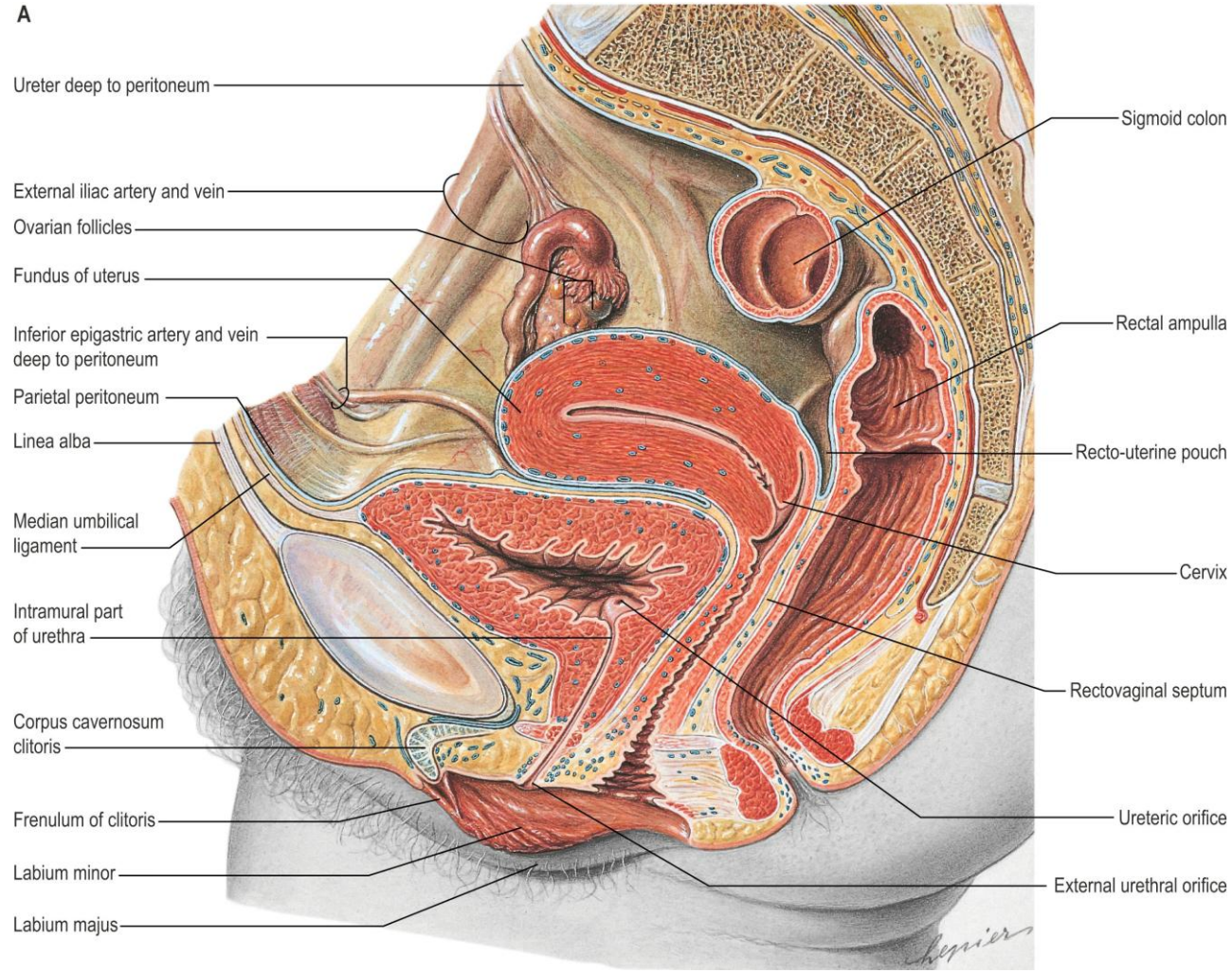


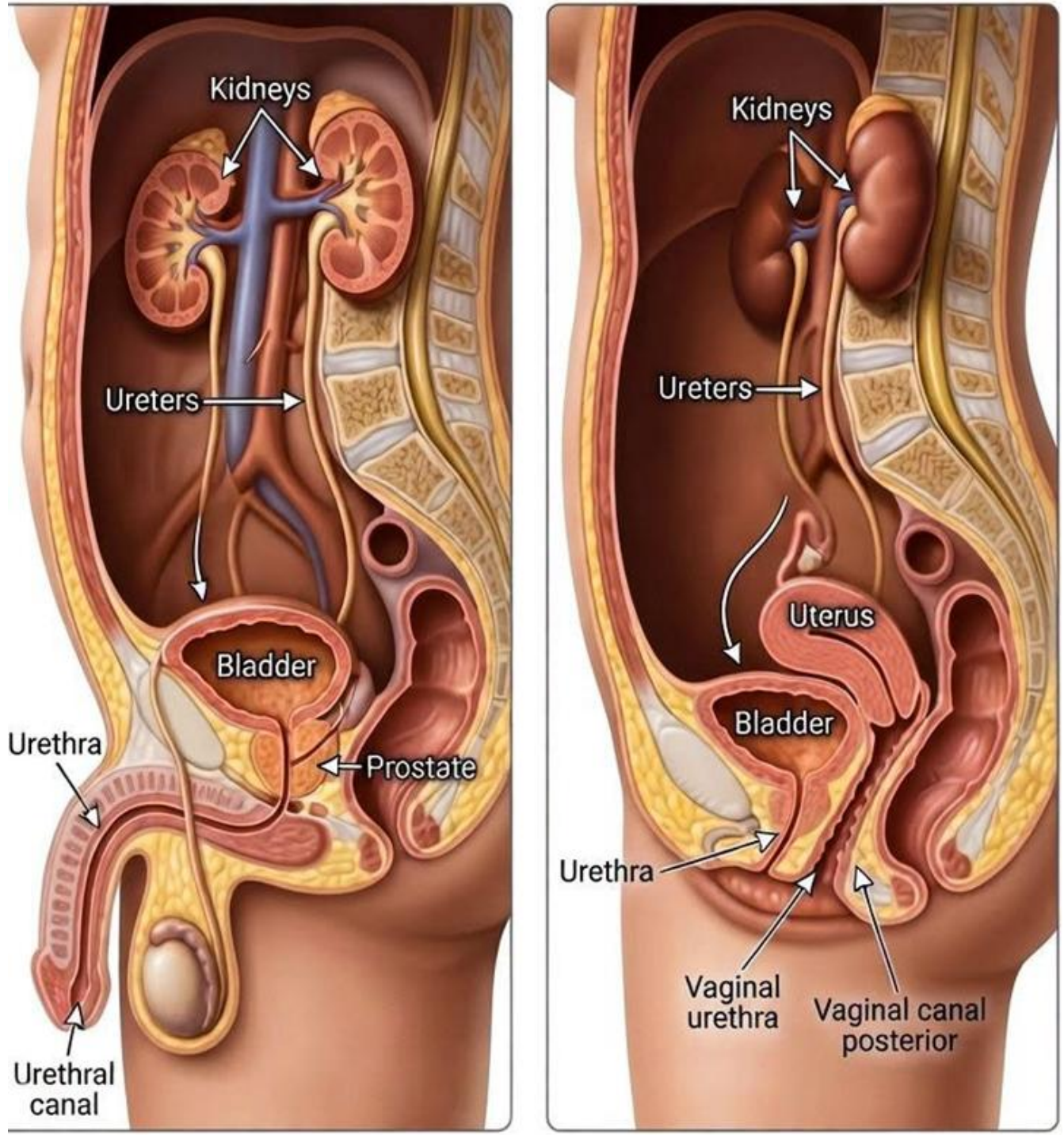
- **Collapsible muscular sac**
- **Stores and expels urine**

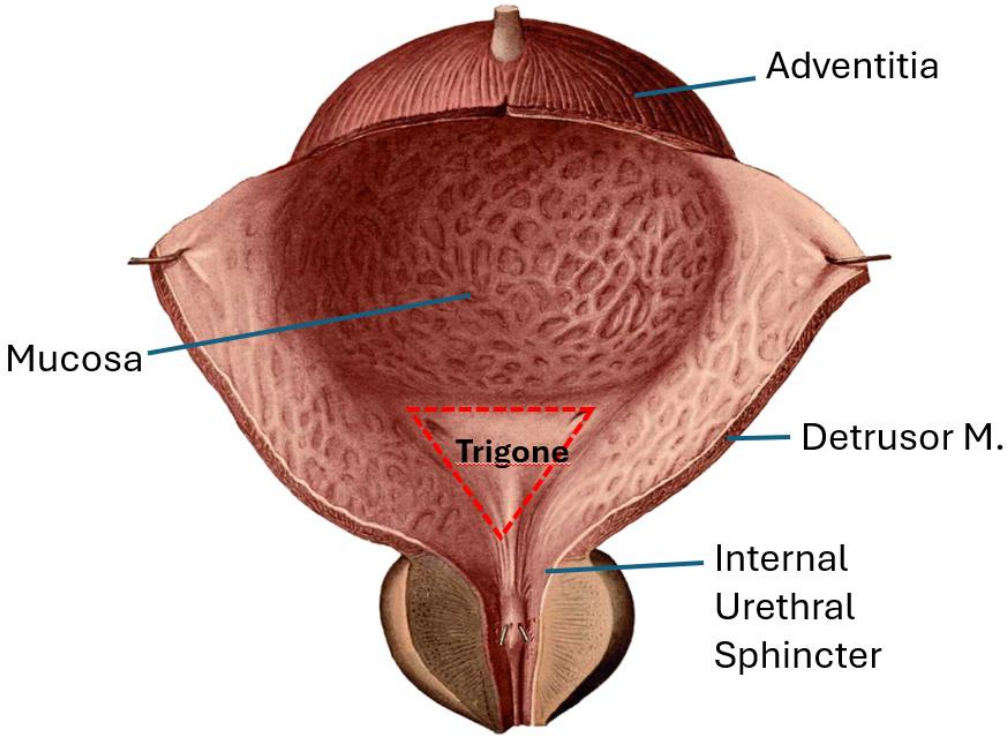
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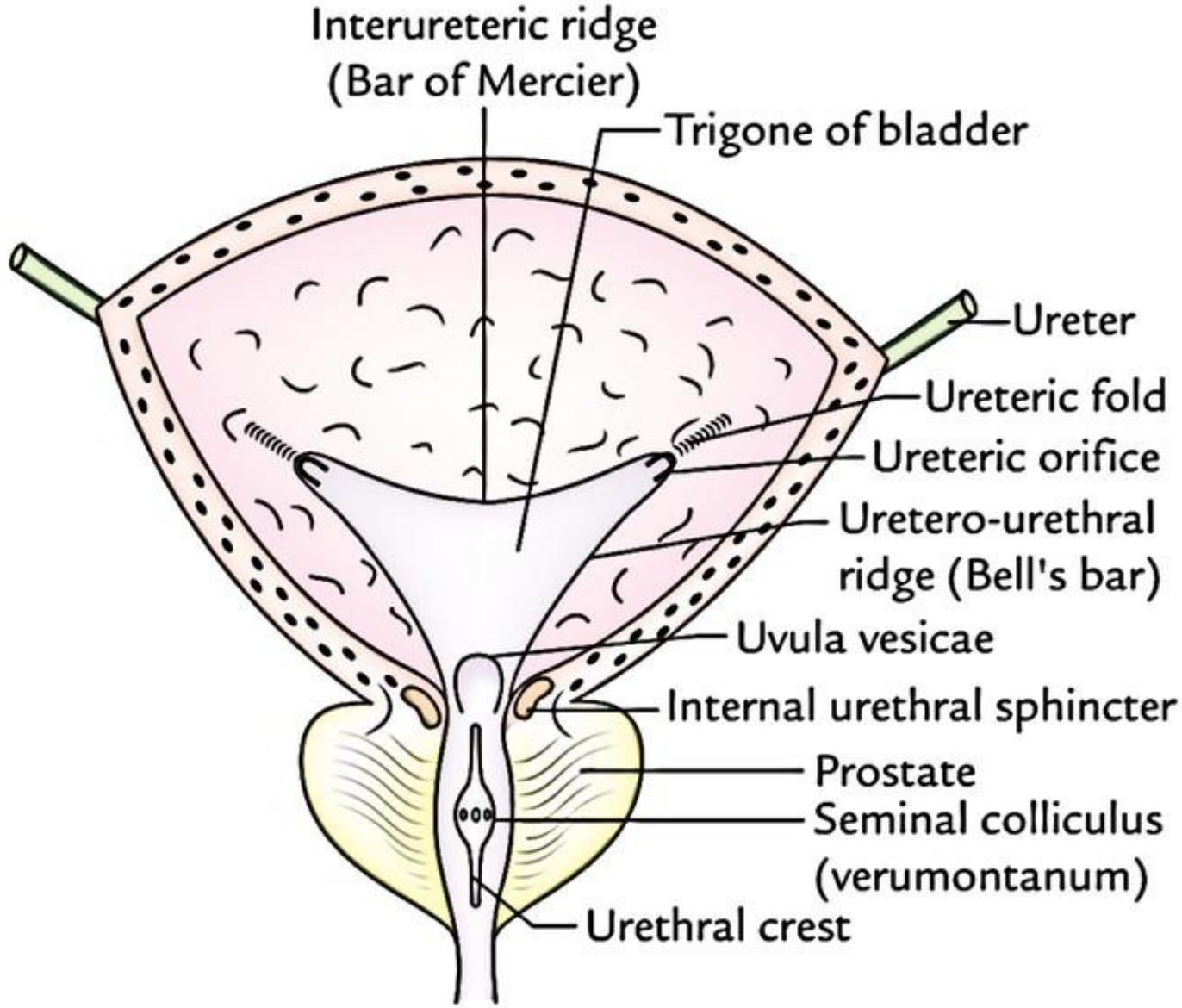

Internal urethral orifice
Small intestine
Median umbilical ligament
Retropubic space (of Retzius)
Linea alba
Pubic symphysis
Suspensory ligament of penis
Deep dorsal vein of penis
Superficial dorsal vein of penis
Sigmoid colon
Ureteric orifice
Rectovesical pouch
Rectal ampulla
Prostato-seminal fascia
Prostate
Visceral pelvic fascia
Anococcygeal ligament
External anal sphincter
Internal anal sphincter
Deep transverse perineal muscle
Perineal membrane
Membranous urethra
Bulb of penis

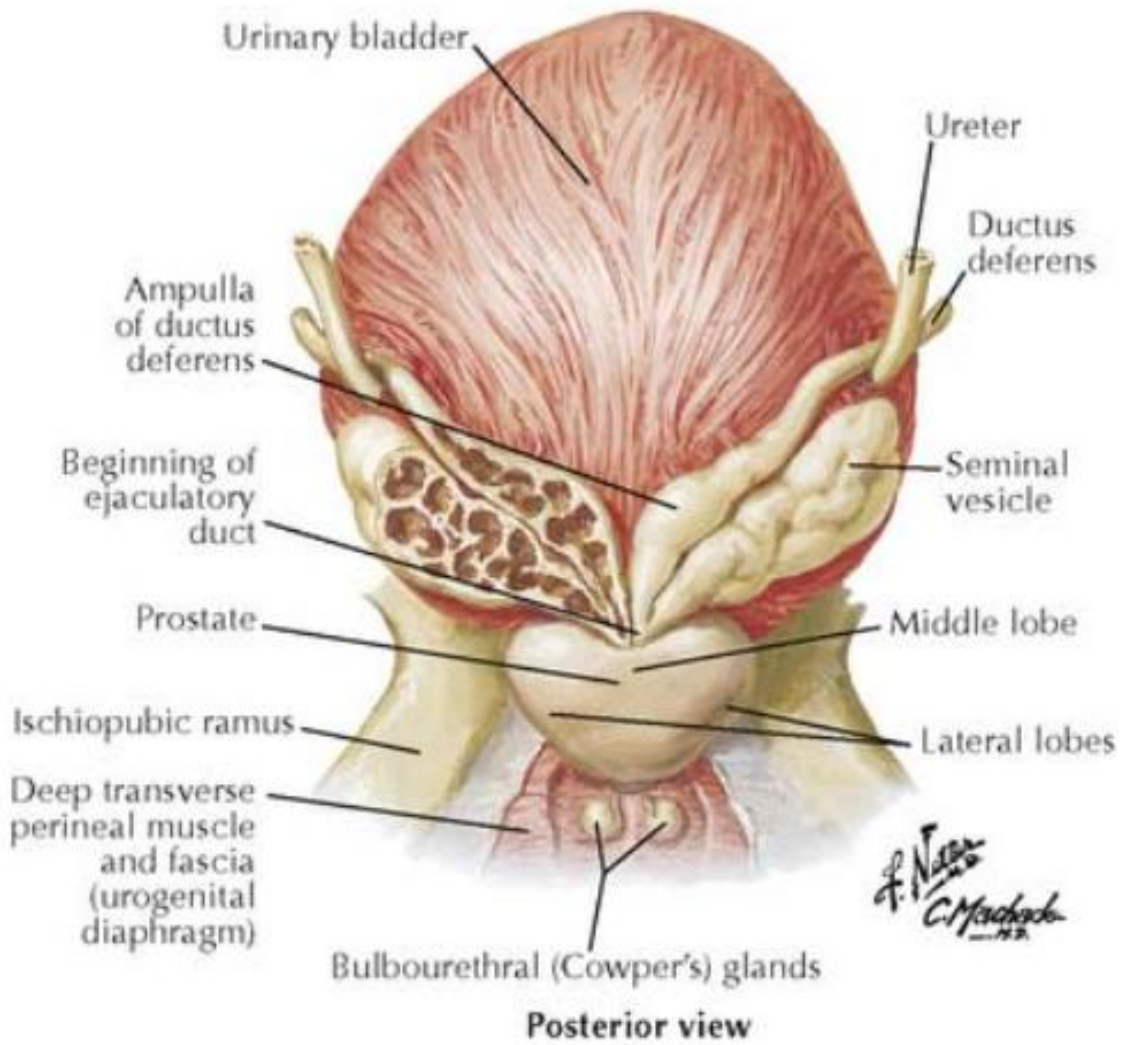
A


Ureter deep to peritoneum
External iliac artery and vein
Ovarian follicles
Fundus of uterus
Inferior epigastric artery and vein deep to peritoneum
Parietal peritoneum
Linea alba
Median umbilical ligament
Intramural part of urethra
Corpus cavernosum clitoris
Frenulum of clitoris
Labium minor
Labium majus
Sigmoid colon
Rectal ampulla
Recto-uterine pouch
Cervix
Rectovaginal septum
Ureteric orifice
External urethral orifice

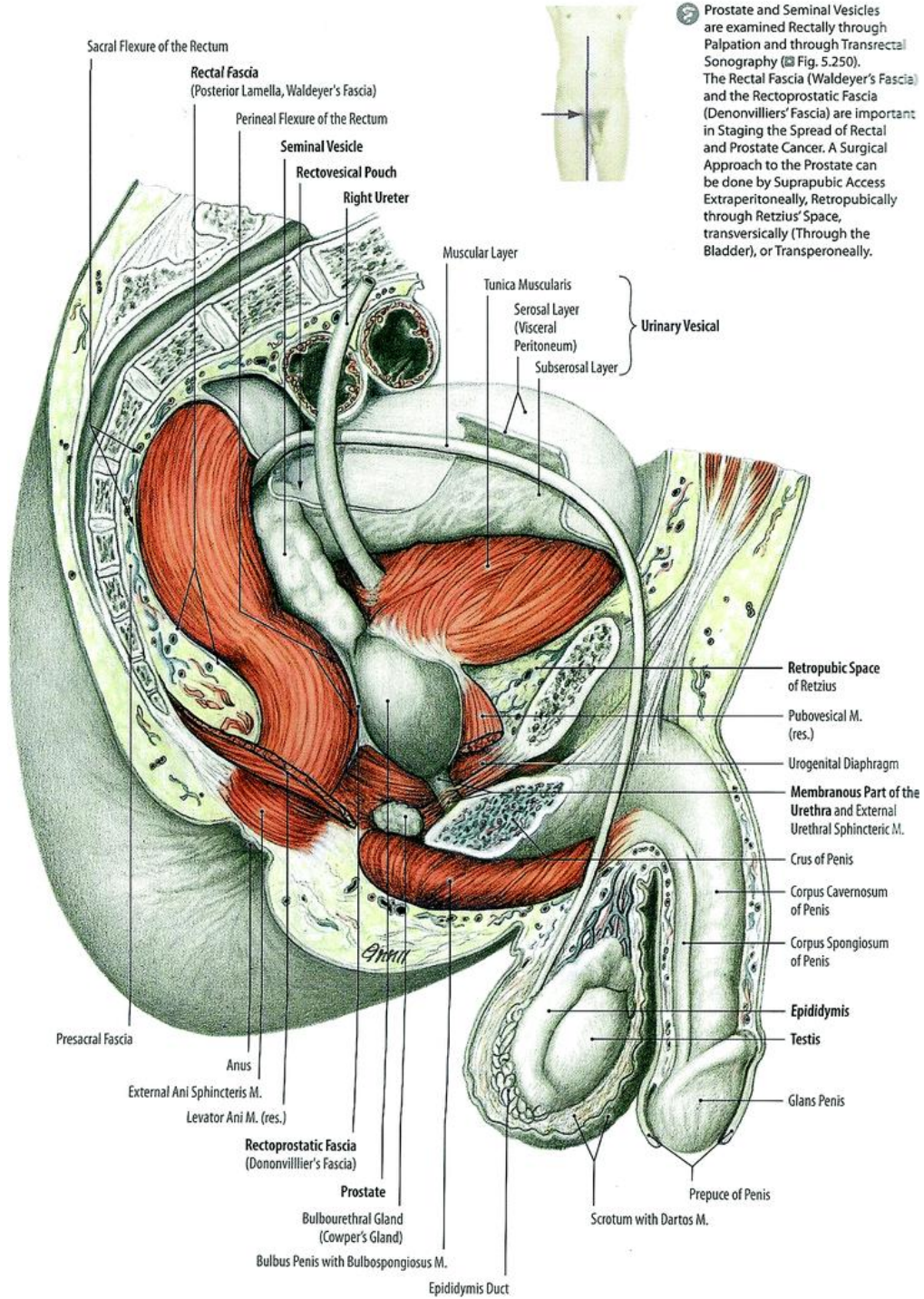

Kidneys
Ureters
Bladder
Urethra
Prostate
Urethral canal
Kidneys
Ureters
Uterus
Bladder
Urethra
Vaginal urethra
Vaginal canal posterior


Adventitia
Mucosa
Trigone
Detrusor M.
Internal
Urethral
Sphincter

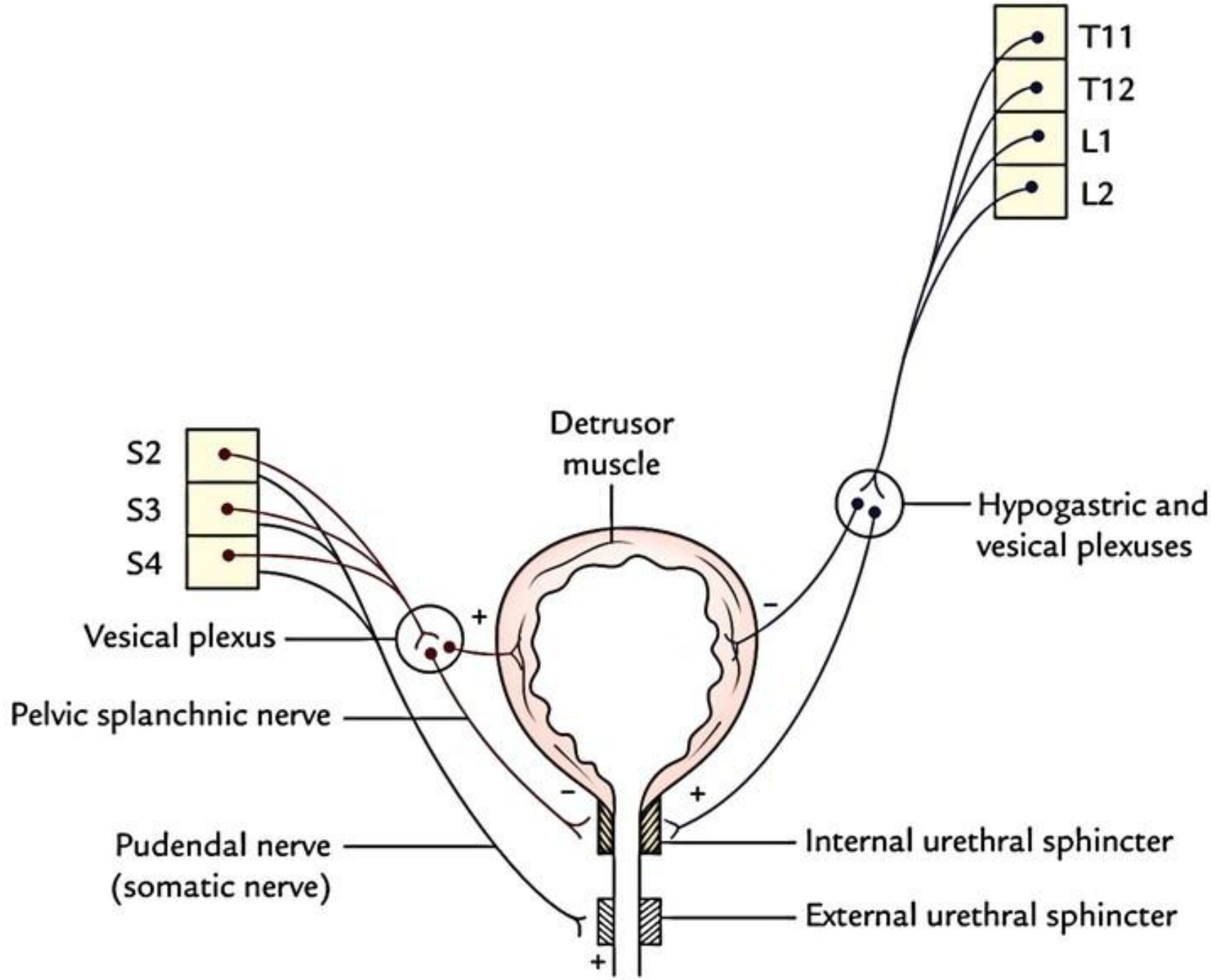

Interureteric ridge
(Bar of Mercier)
Trigone of bladder
Ureter
Ureteric fold
Ureteric orifice
Uretero-urethral
ridge (Bell's bar)
Uvula vesicae
Internal urethral sphincter
Prostate
Seminal colliculus
(verumontanum)
Urethral crest



Posterior view

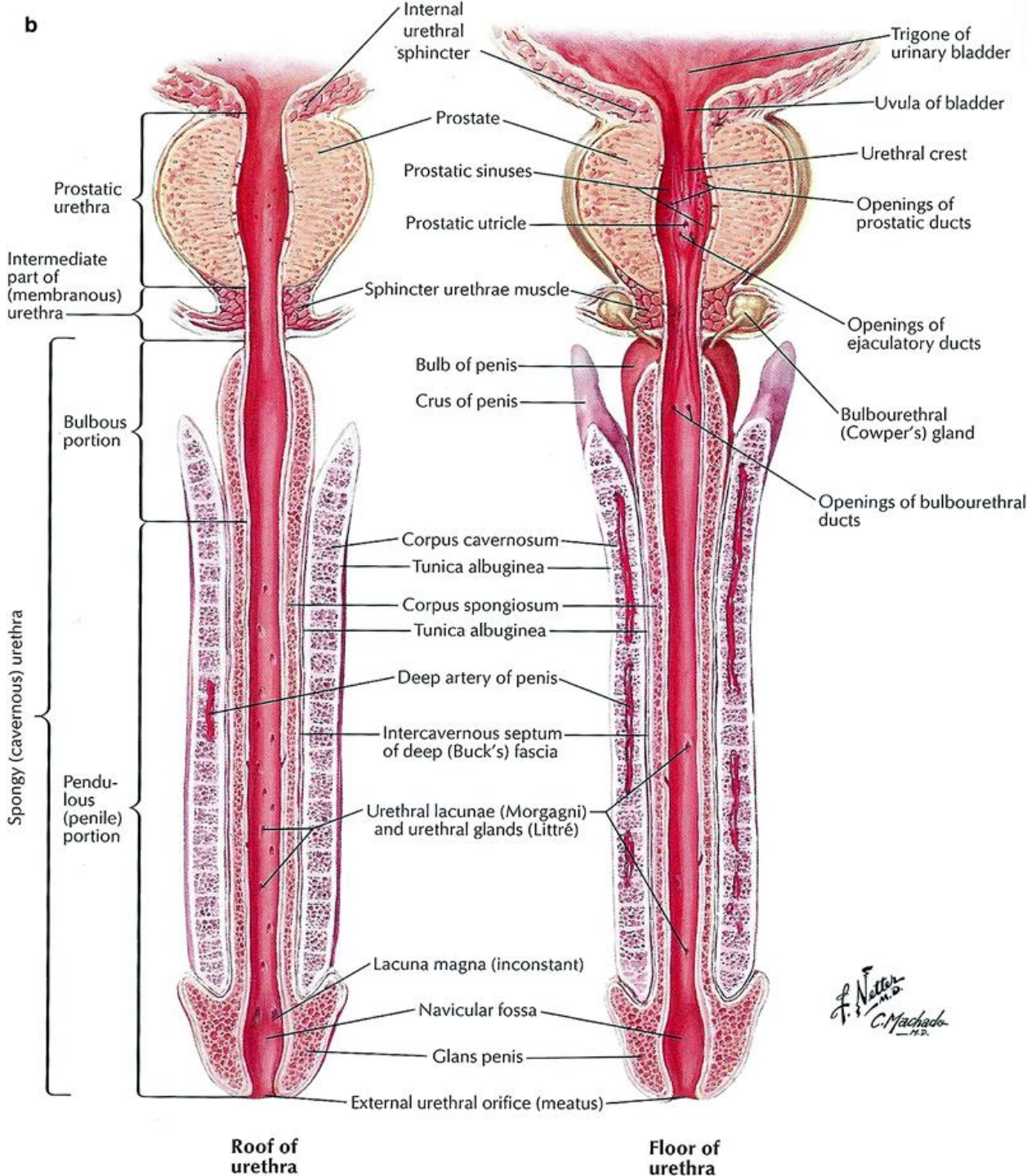


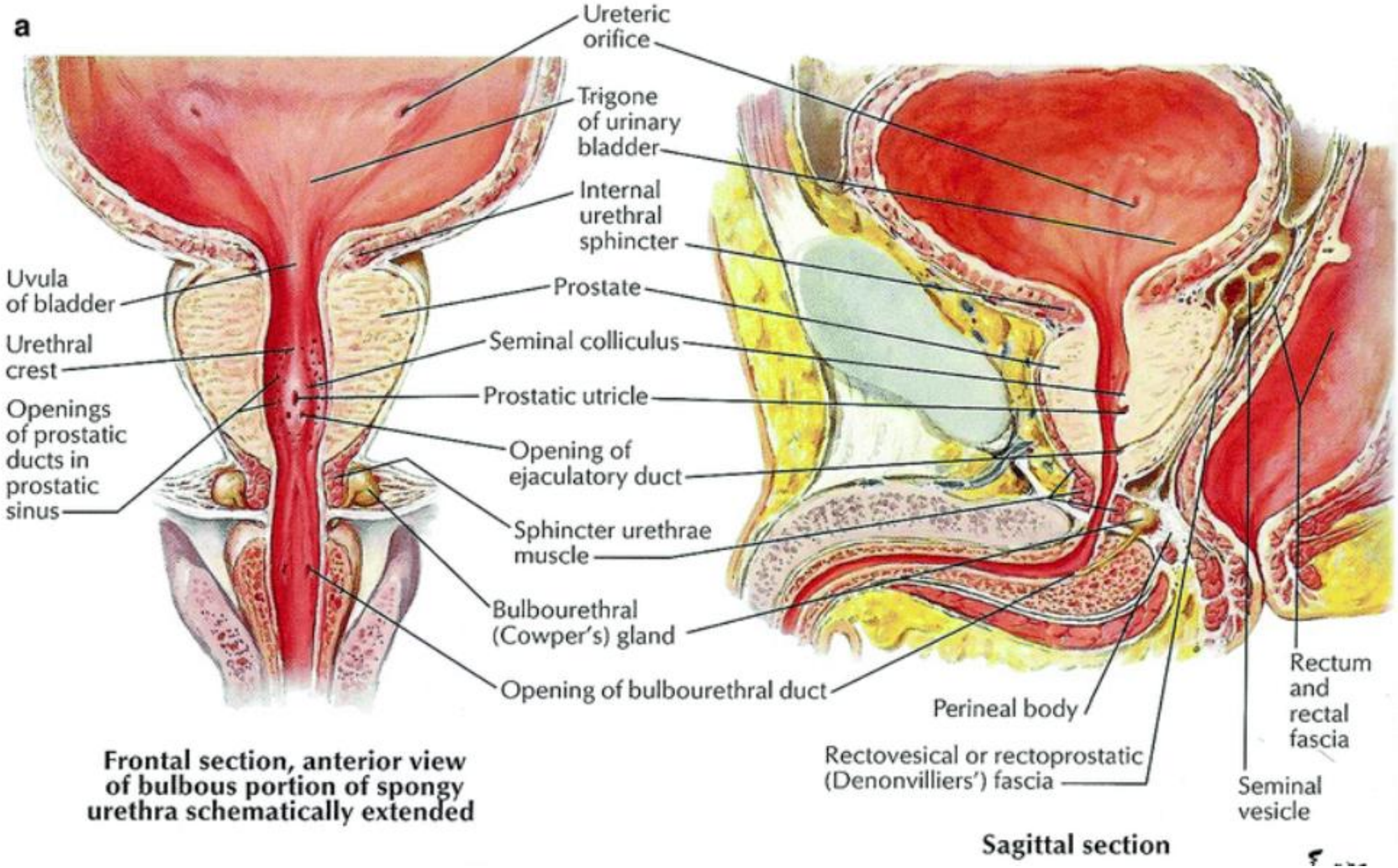
Prostate and Seminal Vesicles are examined Rectally through Palpation and through Transrectal Sonography (Fig. 5.250). The Rectal Fascia (Waldeyer's Fascia) and the Rectoprostatic Fascia (Denonvilliers' Fascia) are important in Staging the Spread of Rectal and Prostate Cancer. A Surgical Approach to the Prostate can be done by Suprapubic Access Extraperitoneally, Retropubically through Retzius' Space, transversically (Through the Bladder), or Transperoneally.

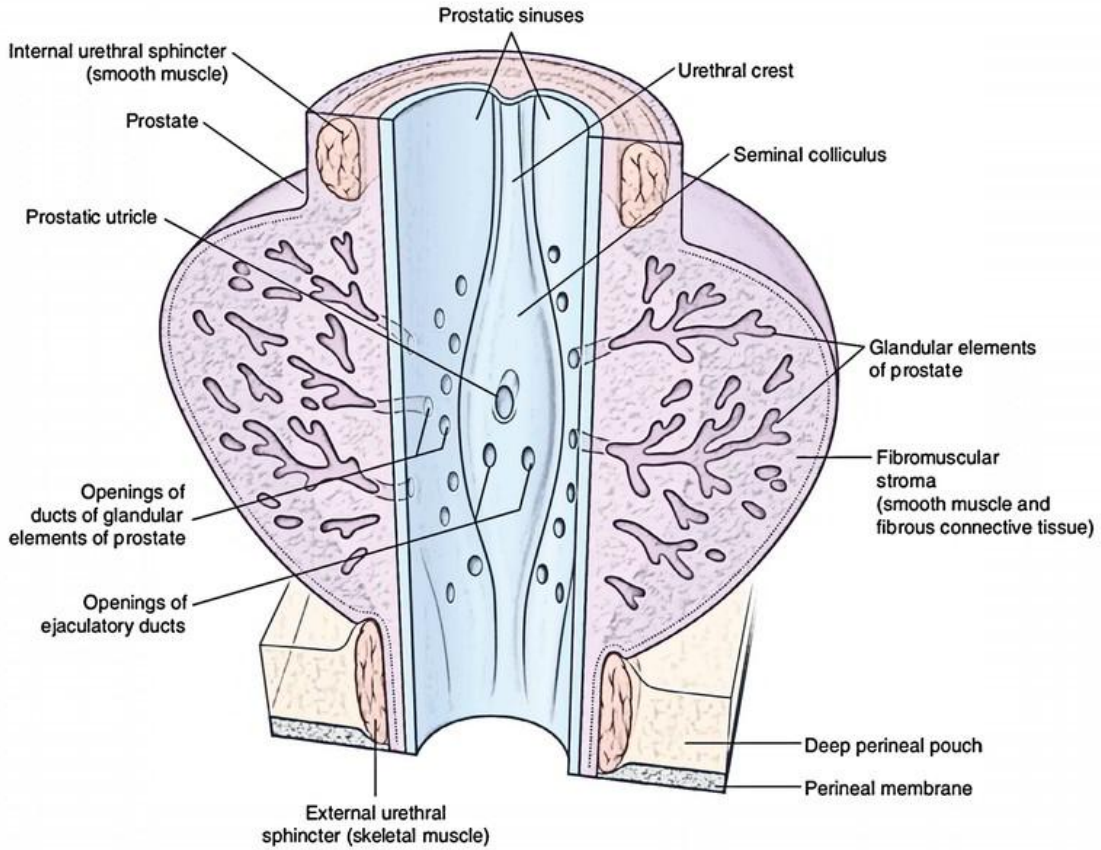

T11
T12
L1
L2
S2
S3
S4
Detrusor muscle
Hypogastric and vesical plexuses
Vesical plexus
Pelvic splanchnic nerve
Pudendal nerve (somatic nerve)
Internal urethral sphincter
External urethral sphincter
+
−
−
+
+

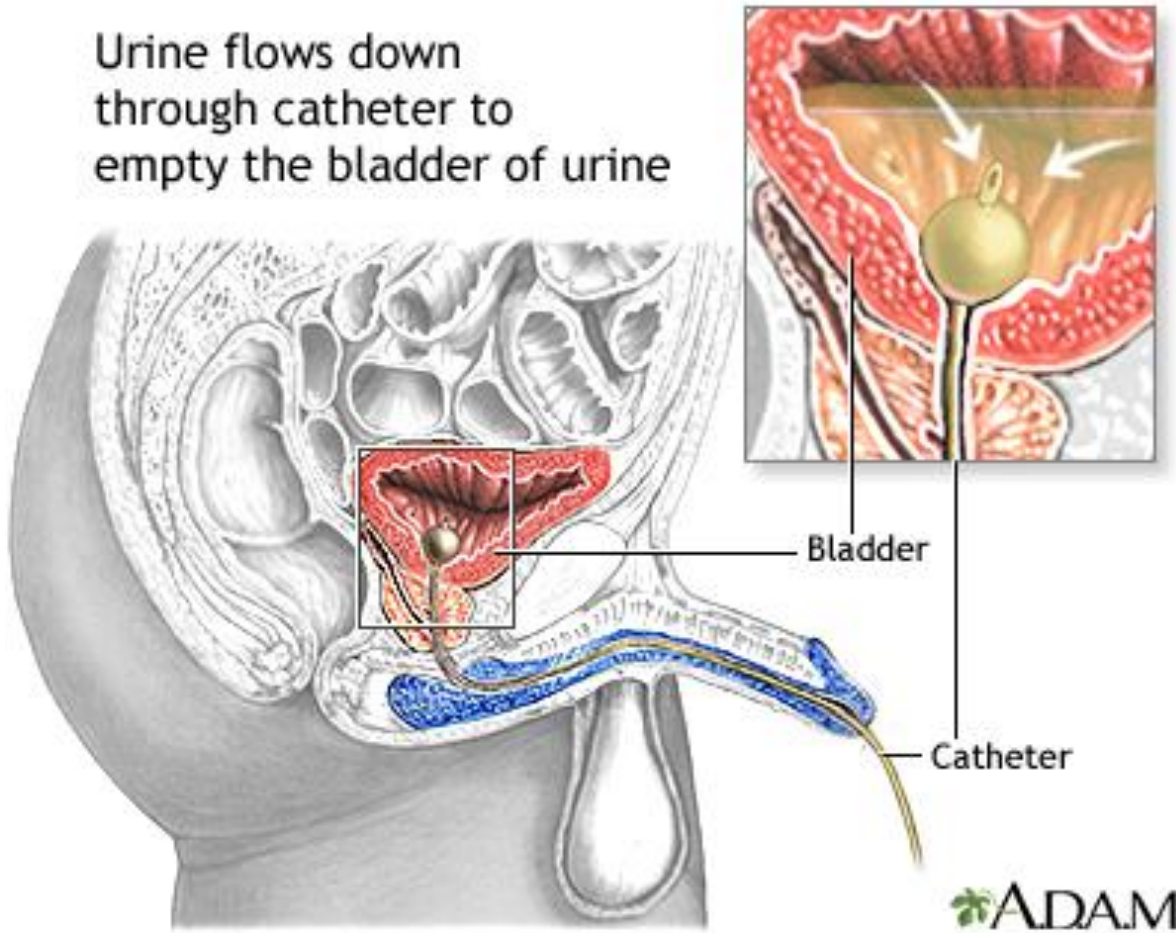
Urethra

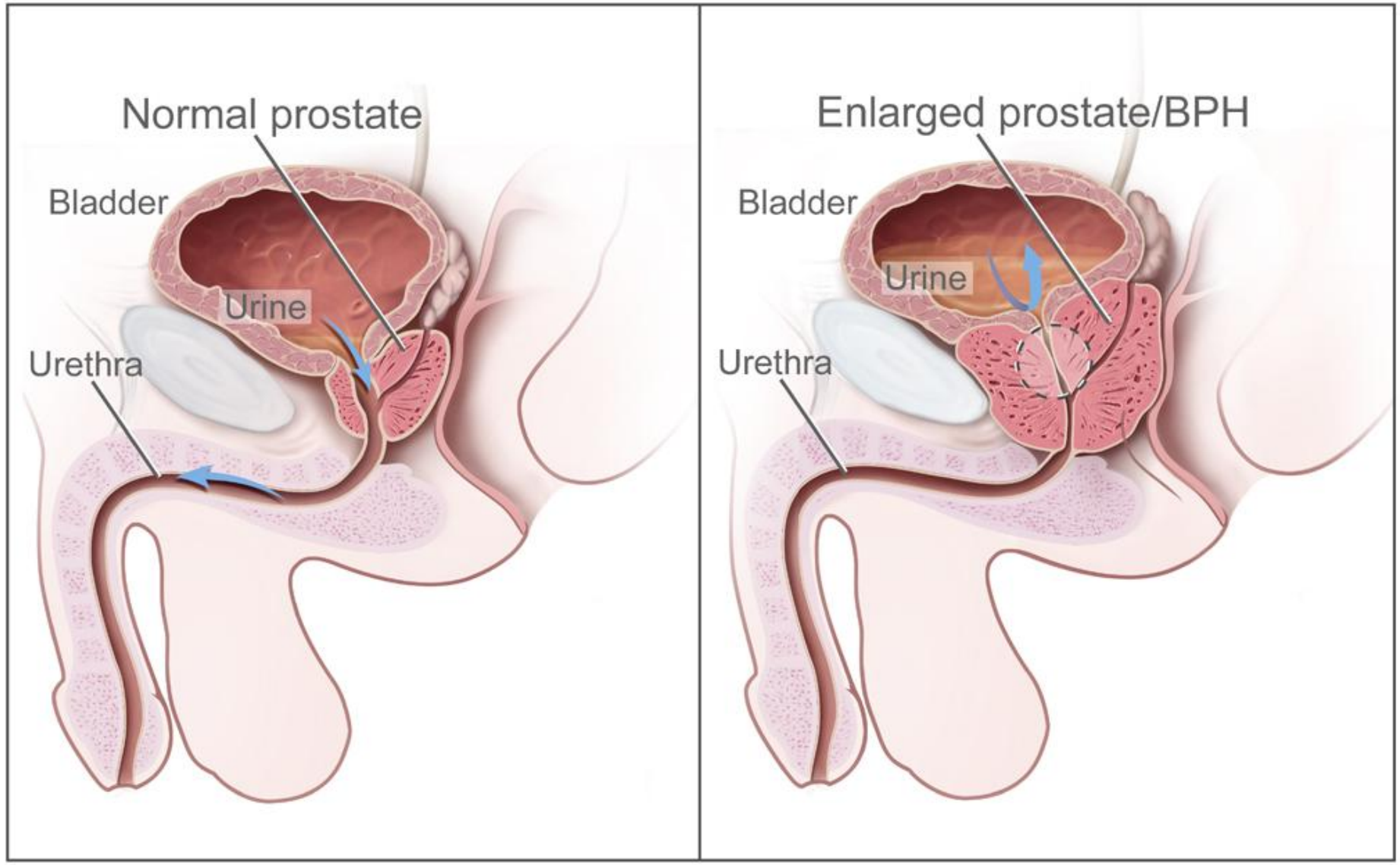
- Extends from the base of the bladder to the outside.
- Anatomical differences mean that male and female urethras are different.
  - Female: 4cm long
  - Male: 14cm long



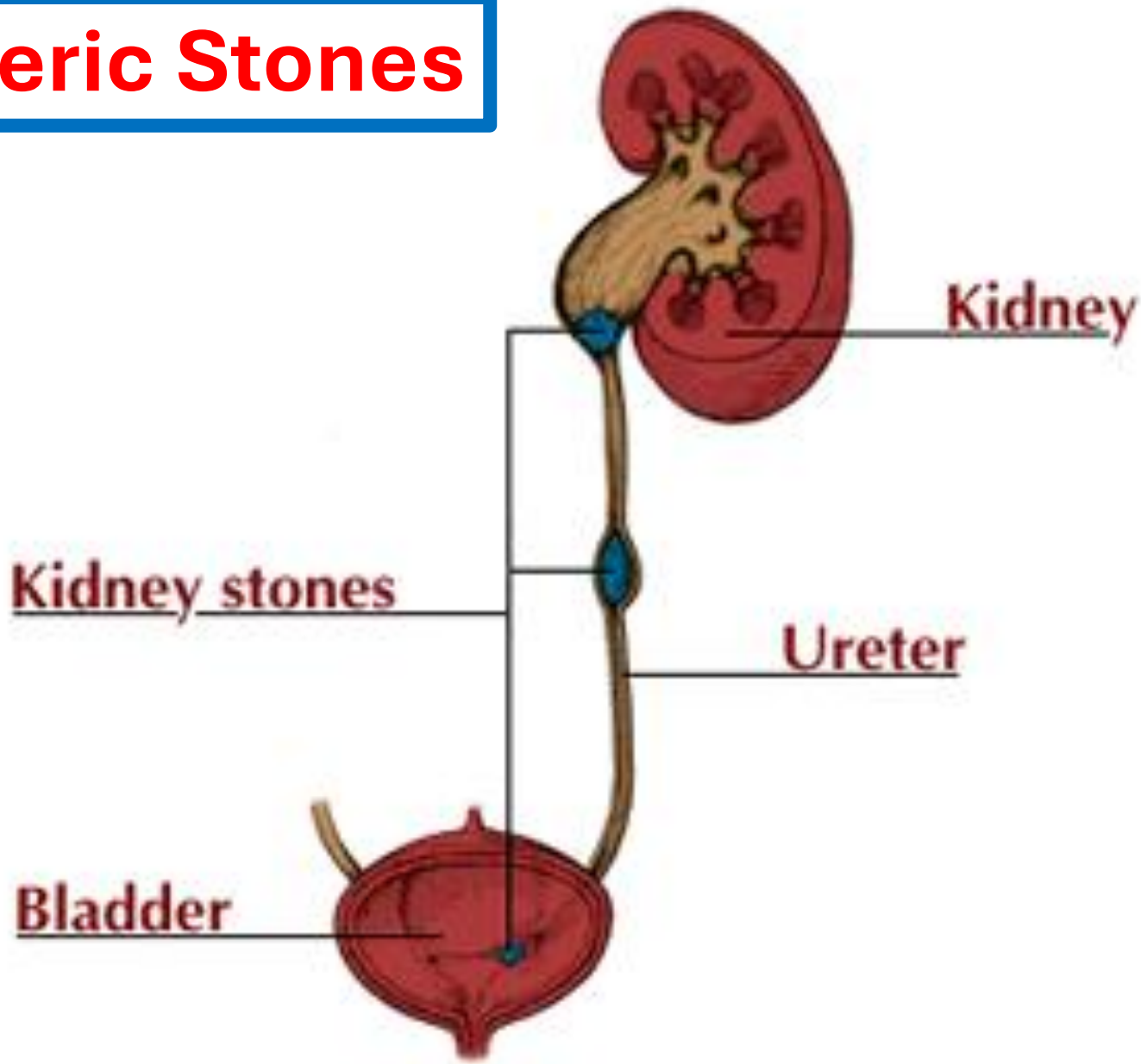

a
Ureteric orifice
Trigone of urinary bladder
Internal urethral sphincter
Uvula of bladder
Prostate
Urethral crest
Seminal colliculus
Openings of prostatic ducts in prostatic sinus
Prostatic utricle
Opening of ejaculatory duct
Sphincter urethrae muscle
Bulbourethral (Cowper's) gland
Opening of bulbourethral duct
Perineal body
Rectum and rectal fascia
Rectovesical or rectoprostatic (Denonvilliers') fascia
Seminal vesicle
Frontal section, anterior view of bulbous portion of spongy urethra schematically extended
Sagittal section

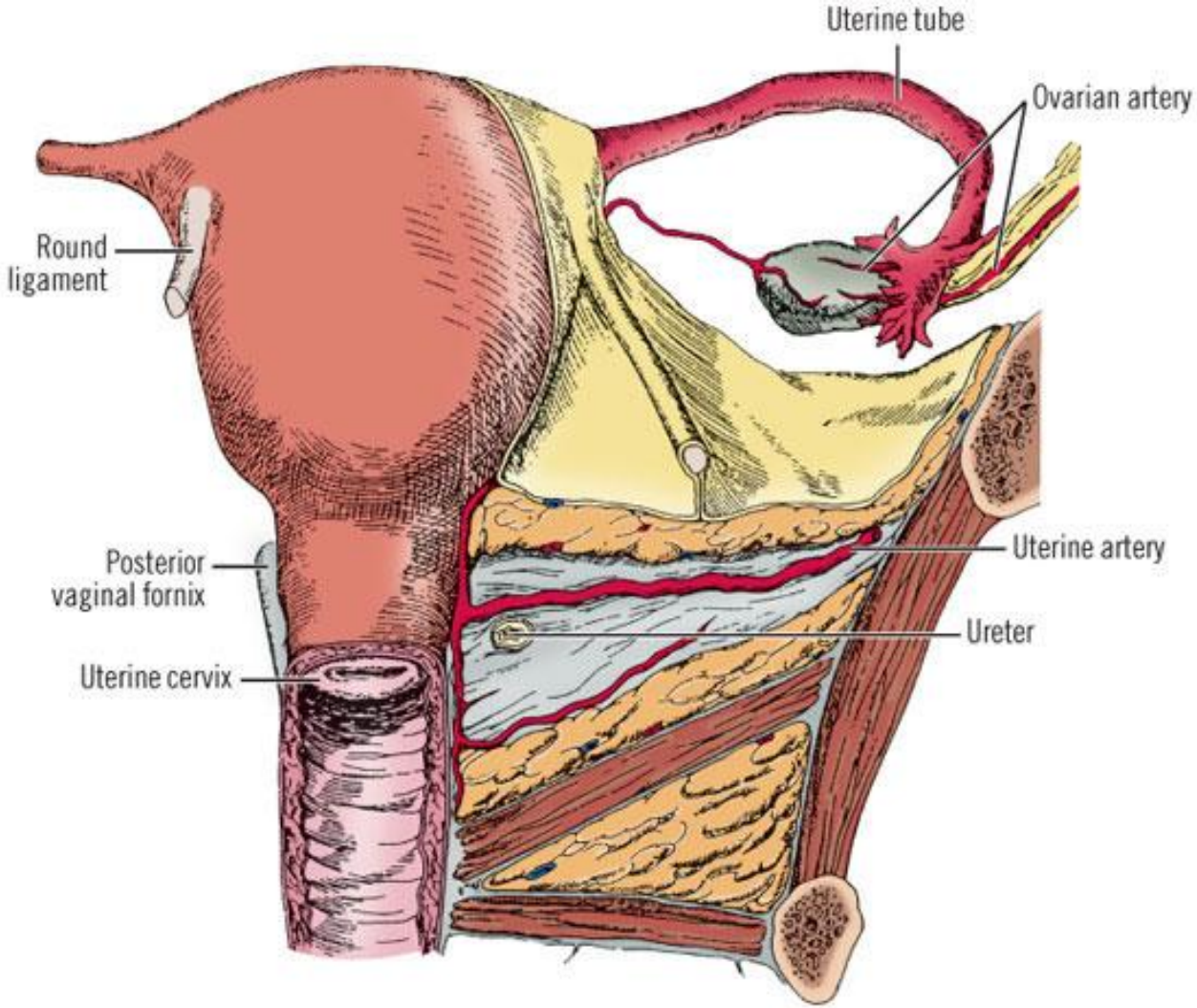

Prostatic sinuses
Internal urethral sphincter (smooth muscle)
Urethral crest
Prostate
Seminal colliculus
Prostatic utricle
Glandular elements of prostate
Fibromuscular stroma (smooth muscle and fibrous connective tissue)
Openings of ducts of glandular elements of prostate
Openings of ejaculatory ducts
Deep perineal pouch
Perineal membrane
External urethral sphincter (skeletal muscle)


Urine flows down through catheter to empty the bladder of urine
Bladder
Catheter
A.D.A.M.

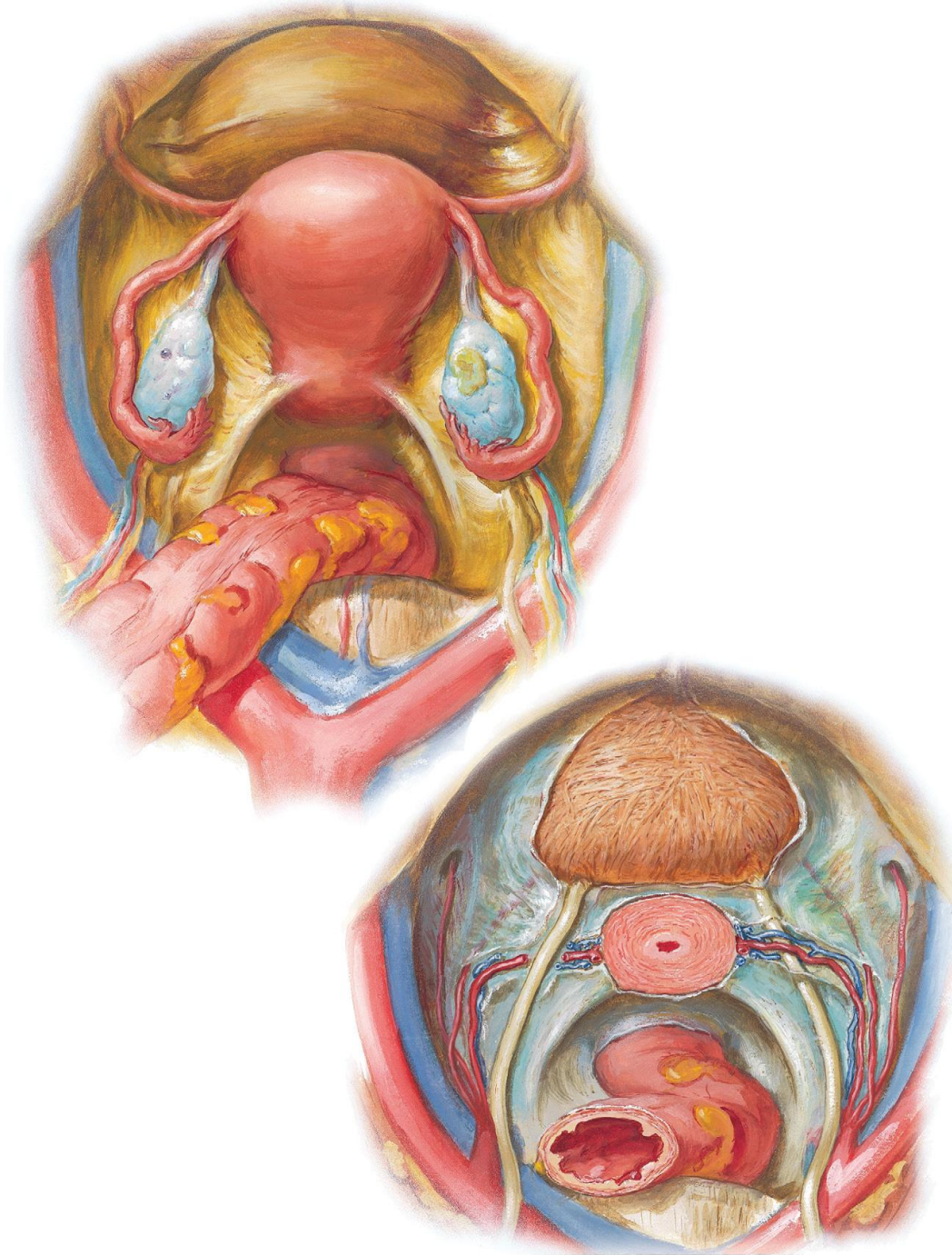

Normal prostate
Bladder
Urine
Urethra
Enlarged prostate/BPH
Bladder
Urine
Urethra

Ureteric Stones





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Scarpa's Fascia

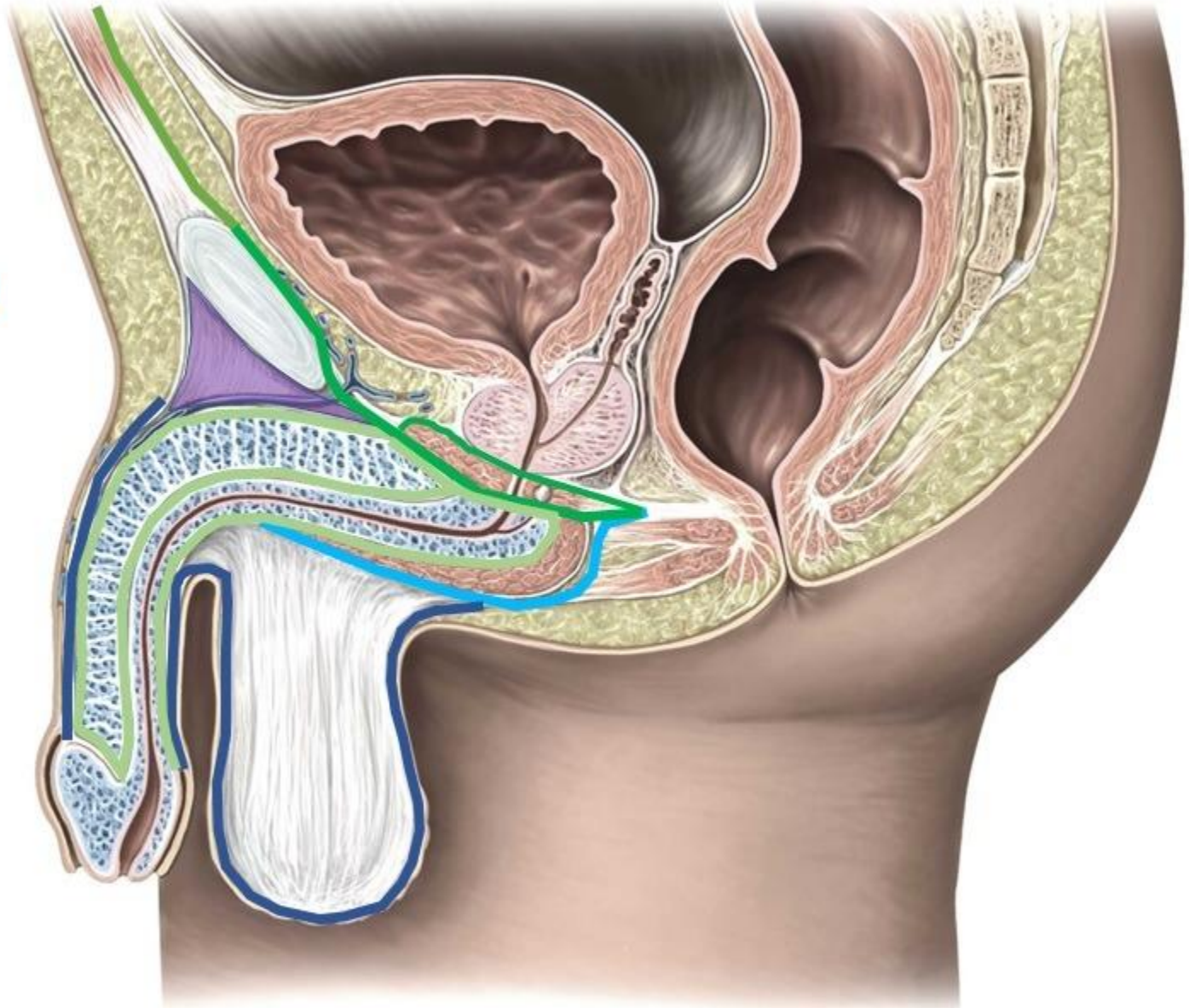
Dartos Fascia

Membranous Perineal (Colle's) Fascia

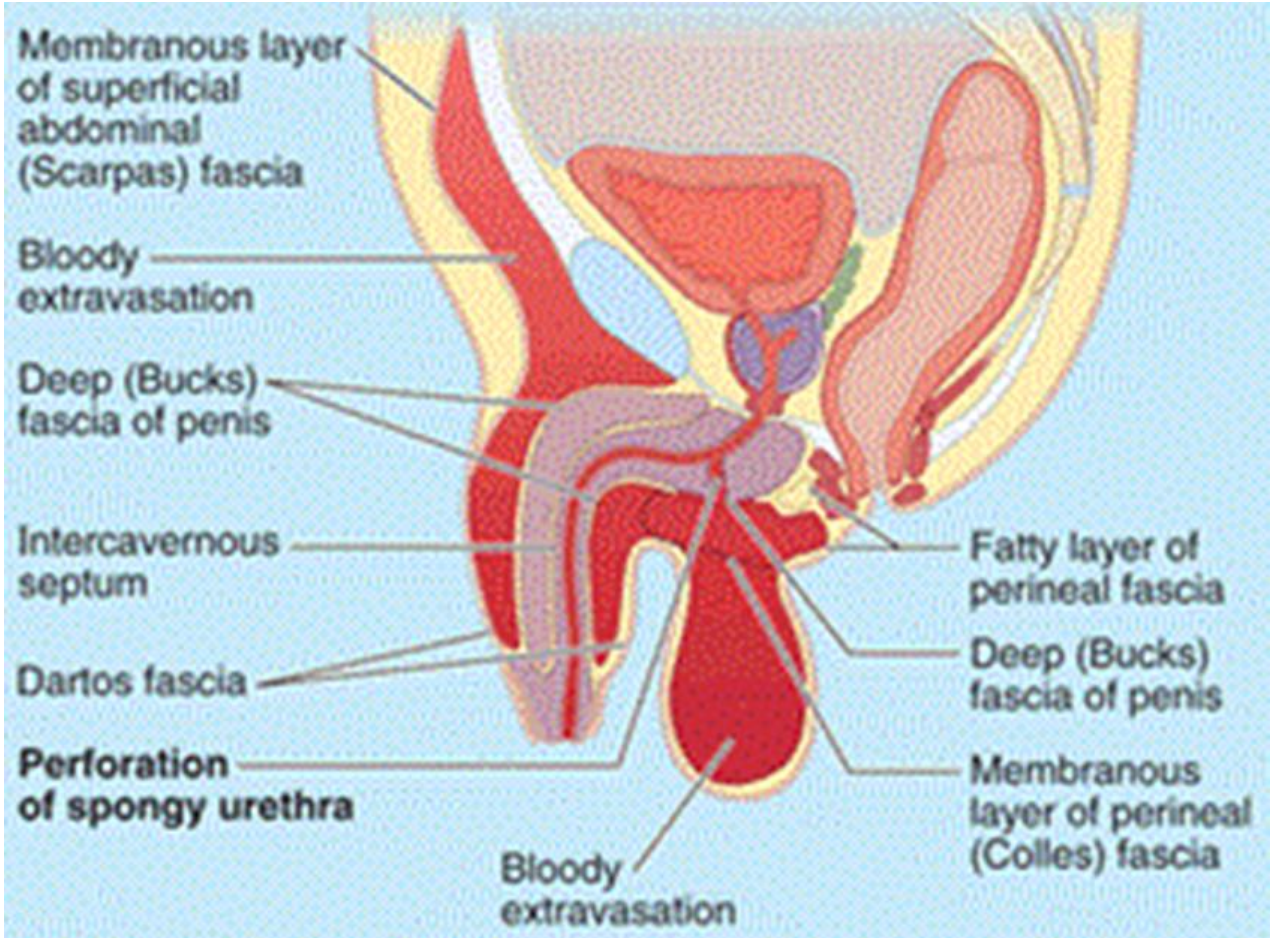
Deep Perineal (Gallaudet) Fascia

Deep (Buck's Fascia of the Penis

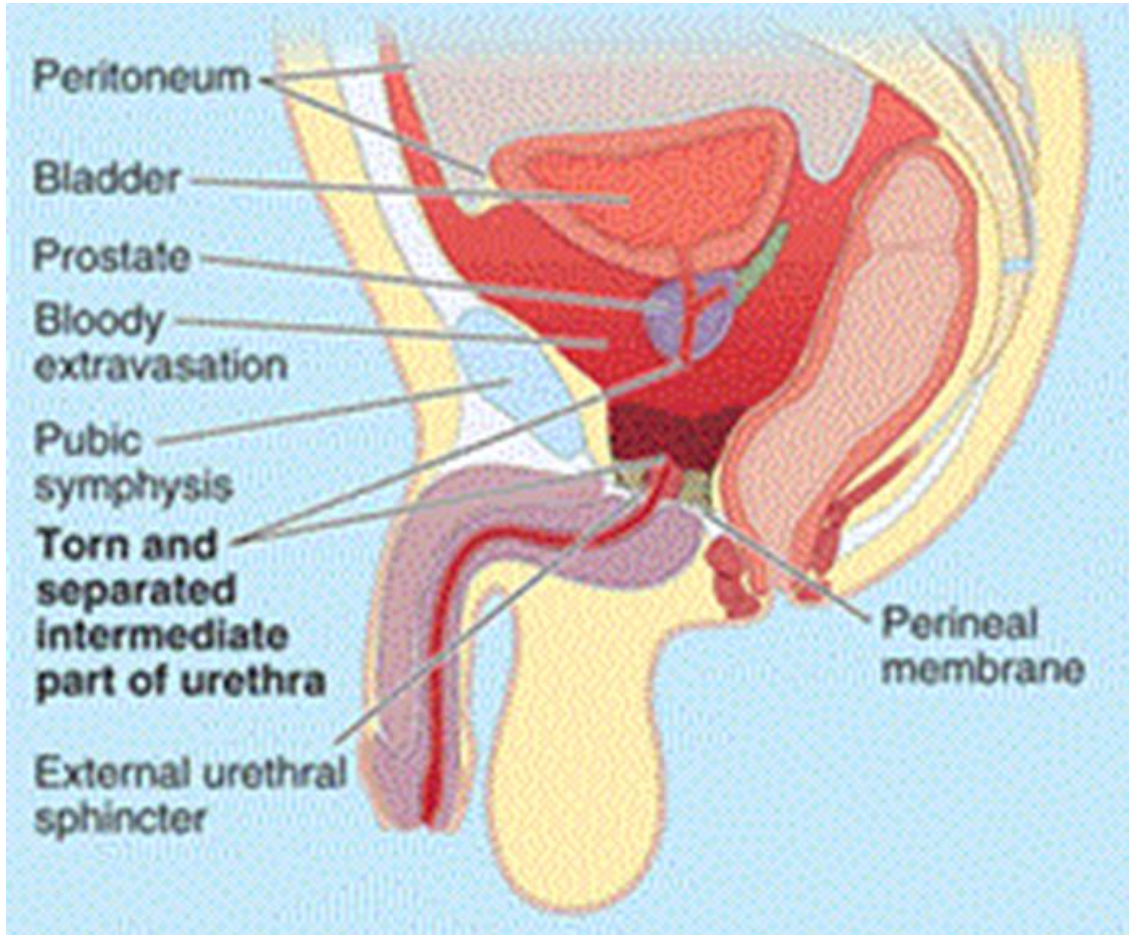
Suspensory Ligament of the Penis



Urethral injury and Urin Extravasation



Anterior urethral injury
(bulbar or penile urethra)
Extra-abdominal



Posterior urethral injury
(membranous or prostatic urethra)
Intra-abdominal / Extraperitoneal

THANKS